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Structure of the Calvin-Benson-Bassham sedoheptulose-1,7-bisphosphatase from the model microalga *Chlamydomonas reinhardtii*

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Abstract

The Calvin-Benson-Bassham cycle (CBBC) performs carbon fixation in photosynthetic organisms. Among the eleven enzymes that participate in the pathway, sedoheptulose-1,7-bisphosphatase (SBPase) is expressed in photo-autotrophs and catalyzes the hydrolysis of sedoheptulose-1,7-bisphosphate (SBP) to sedoheptulose-7-phosphate (S7P). SBPase, along with nine other enzymes in the CBBC, contributes to the regeneration of ribulose-1,5-bisphosphate, the carbon-fixing co-substrate used by ribulose-1,5-bisphosphate carboxylase/oxygenase (Rubisco). The metabolic role of SBPase is restricted to the CBBC, and a recent study revealed that the three-dimensional structure of SBPase from the moss *Physcomitrium patens* was found to be similar to that of fructose-1,6-bisphosphatase (FBPase), an enzyme involved in both CBBC and neoglucogenesis. In this study we report the first structure of an SBPase from a chlorophyte, the model unicellular green microalga *Chlamydomonas reinhardtii*. By combining experimental and computational structural analyses, we describe the topology, conformations and quaternary structure of *Chlamydomonas reinhardtii* SBPase (CrSBPase). We identify active site residues and locate sites of redox- and phospho-post-translational modifications that contribute to enzymatic functions. Finally, we observe that CrSBPase adopts distinct oligomeric states that may dynamically contribute to the control of its activity.

eLife assessment

The manuscript reports **valuable** findings by resolving the crystal structure of Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae *Chlamydomonas reinhardtii*, which is involved in the Calvin cycle. The data presented are **solid**, based on validated methodologies, and they help in understanding the structure and function of this enzyme.

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Introduction

Photosynthetic carbon fixation is performed by eleven enzymes that collectively operate the Calvin-Benson-Bassham cycle (CBBC) (Calvin 1962 [↗](#), Johnson 2016 [↗](#)), wherein Rubisco plays the prominent role in catalyzing fixation of inorganic carbon onto a pentose acceptor: ribulose-1,5-bisphosphate (RuBP). The net gain of the CBBC is one triose in the form of glyceraldehyde-3-phosphate (G3P) for every three carboxylation reactions catalyzed by Rubisco (Johnson and Alric 2013 [↗](#)). Regeneration of RuBP is ensured by the other ten enzymes of the CBBC, among which phosphoribulokinase (PRK) and sedoheptulose-1,7-bisphosphatase (SBPase) are the only enzymes that are exclusively involved in this pathway. The remaining eight enzymes are paralogs of those involved in cytoplasmic glycolysis, gluconeogenesis or the pentose-phosphate pathway. Hence, Rubisco, PRK, and SBPase are expressed in photosynthetic organisms and fulfill biochemical functions so far uniquely attributed to the photosynthetic carbon fixation.

Pioneering studies on SBPase were carried out using the native enzyme purified from wheat chloroplasts (Woodrow 1982 [↗](#)) and corn leaves (Nishizawa and Buchanan 1981 [↗](#)), allowing quantitative assays of its catalytic function. This enzyme catalyzes the hydrolysis of sedoheptulose-1,7-bisphosphate (SBP) into sedoheptulose-7-phosphate (S7P) and requires magnesium ions for both catalysis and activation. In addition, SBPase activity is modulated by pH and redox switching mechanisms (Anderson 1974 [↗](#), Woodrow and Walker 1980 [↗](#), Cadet and Meunier 1988 [↗](#)). Redox control is exerted through dithiol/disulfide interchanges dependent upon the ferredoxin-thioredoxin cascade arising from the activity of the illuminated photosystems (Schürmann and Buchanan 1975 [↗](#)). In tobacco leaf extracts, SBPase reaches its maximal activity after 10 minutes illumination (Zimmer, Swart et al. 2021 [↗](#)), and this activation kinetics is consistent with the slow reduction pattern of *Arabidopsis* SBPase *in vitro* (Yoshida and Hisabori 2018 [↗](#)).

The crystal structure of SBPase from the moss *Physcomitrium patens* (PpSBPase) has been solved at a resolution of 1.3 Å (Protein Data Bank entry : 5iz3) (Güttele, Roret et al. 2016 [↗](#)). PpSBPase fold is highly similar to that of moss fructose-1,6-bisphosphatase (FBPase), yet the protein was shown to distinctively oligomerize as a homodimer and to form a disulfide bridge in a different motif than that found in plant FBPase (Chiadmi, Navaza et al. 1999 [↗](#)) (Güttele, Roret et al. 2016 [↗](#)). In spite of their homology, the sequences of FBPase and SBPase seem to have evolved different regulatory and catalytic properties that allow different contributions to the CBBC. Indeed, both enzymes specifically recognize their respective 6-carbon and 7-carbon sugar substrates (i.e., fructose-1,6-bisphosphate and SBP, respectively), with only SBPase capable to hydrolyze the other substrate with a resulting catalysis lower than that measured with SBP (Güttele, Roret et al. 2016 [↗](#)). To date, the structural components contributing to active site selectivity (i.e., substrate recognition) can be retrieved from two studies reporting the crystal structure of the dual function fructose-1,6-bisphosphatase/sedoheptulose-1,7-bisphosphatase (F/SBPase) from *Thermosynechococcus elongatus* in complex with SBP (Cotton, Kabasakal et al. 2015 [↗](#)) and from *Synechocystis* sp. PCC

6803 in complex with FBP (PDB entry 3RPL). Recently, it was proposed that the CBBC can bypass fructose-1,6-bisphosphate (FBP) altogether, by only relying on SBP, glycolysis and the oxidative pentose-phosphate pathway paralogs to regenerate the substrate of Rubisco (Ohta 2022 [\[1\]](#)).

Several studies have reported that SBPase, along with Rubisco, FBPase, aldolase and transketolase, constitute a metabolic bottleneck for the CBBC (Raines 2022 [\[2\]](#) and references therein). Overexpression of SBPase was found to effectively enhance photosynthesis and growth in plants (Lefebvre, Lawson et al. 2005 [\[3\]](#), Feng, Han et al. 2007 [\[4\]](#), Feng, Wang et al. 2007 [\[5\]](#), Rosenthal, Locke et al. 2011 [\[6\]](#), Simkin, Lopez-Calcano et al. 2017 [\[7\]](#)) and in the green microalga *Chlamydomonas reinhardtii* (Hammel, Sommer et al. 2020 [\[8\]](#)), indicating that SBPase can be considered as a limiting factor for the photosynthetic carbon fixation. Based on these evidences, it is plausible to consider that SBPase is a good target to improve the photosynthetic capacity by means of metabolic engineering coupled to synthetic biology (Kubis and Bar-Even 2019 [\[9\]](#)), especially in the experimentally convenient multi-omic model alga *Chlamydomonas* (Mettler, Muhlhaus et al. 2014 [\[10\]](#), Schmollinger, Muhlhaus et al. 2014 [\[11\]](#)).

In *Chlamydomonas*, SBPase is encoded by a single nuclear gene (Cre03.g185550), and the protein (Uniprot entry P46284) is addressed to the chloroplast by a 54-residue amino-terminal peptide (Emanuelsson, Brunak et al. 2007 [\[12\]](#), Almagro Armenteros, Salvatore et al. 2019 [\[13\]](#)). SBPase abundance in the algal cell was quantified by mass spectrometry as $1.1\text{--}1.3 \pm 0.3$ amol/cell, representing 0.15% of total cell proteins (Hammel, Sommer et al. 2020 [\[8\]](#)). In *Chlamydomonas* protein extracts, SBPase was shown to undergo post-translational modifications by phosphorylation of serines and threonines (Werth, McConnell et al. 2017 [\[14\]](#)), and redox modifications of cysteine thiols (Zaffagnini, Bedhomme et al. 2012 [\[15\]](#), Morisse, Zaffagnini et al. 2014 [\[16\]](#), Pérez-Pérez, Mauriès et al. 2017 [\[17\]](#)). However, the mechanisms by which post-translational modifications affect the catalytic activity of CrSBPase is still lacking. Here, we describe the crystal structures of CrSBPase in two redox states, examine the catalytic activity of recombinant CrSBPase, and investigate the correlation between the redox state and the oligomerization of the protein. Our results suggest that a redox control over oligomer equilibrium could represent a novel mechanism to regulate enzymatic activities in the CBBC.

Material and methods

Chemical reagents were purchased from Merck (Darmstadt, Germany), chromatography material was provided by Cytiva (Vélizy-Villacoublay, France), crystallization solutions and consumable were purchased from Qiagen (Hilden, Germany), Hampton Research (Aliso Viejo, CA USA), and SPT Labtech (Melbourn, UK).

Cloning and mutagenesis of CrSBPase expression plasmids

The sequence coding for full-length mature SBPase was amplified by reverse transcriptase polymerase chain reaction (RT-PCR) from *Chlamydomonas reinhardtii* total ribonucleic acids extract and inserted at sites 5'-NdeI and BamHI-3' into expression vector pET3c modified for the fusion of an amino-terminal hexa-histidine tag in the recombinant protein, yielding plasmid pSL-175-His₆-CrSBPase-WT. PCR primers were 5'-TCTCGCATATGGCCGTTCTGACCCAGGCC-3' (forward, NdeI site underlined) and 5'-CGGGTGGGATCCTTAGGCAGCCACCTTCTC-3' (reverse, BamHI site underlined). The encoded protein encompasses 331 amino acids, with a molar mass of 36,176.7 Da and a theoretical extinction coefficient of $24,722.5 \text{ M}^{-1} \cdot \text{cm}^{-1}$. Unless otherwise stated, amino acid residues numbering is that of GenBank entry DAA79954.1. Point mutants of cysteines 115 and 120 singly substituted with serines were generated by site-directed PCR mutagenesis with primers listed in **supplementary table 1** [\[18\]](#), producing plasmids pSL-181-His₆-CrSBPase-C115S, and pSL-182-His₆-CrSBPase-C120S.

Recombinant CrSBPase purification

Plasmids pSL-175, -181, and -182 were used to transform *Escherichia coli* BL21(DE3). The transformants were grown at 37 °C in 2YT supplemented with 150 µg/mL ampicillin. When cultures reached an OD₆₀₀ of ~0.6, T7-dependent CrSBPase overexpression was induced by the addition of isopropyl-β-D-thiogalactopyranoside (IPTG) at 0.2 mM for 3 h.

Harvested cell pellets were resuspended in 20 mM Tris pH 7.9, 100 mM NaCl (buffer A) and lysed by 1 sec / 1 sec pulsed sonication for 3 min. The soluble fraction of the lysate was separated by 20 min centrifugation at 20,000 *g* and loaded on 2 mL Ni-NTA resin for His-tag mediated affinity chromatography. The resin was washed in four steps with 25 mL buffer A, 25 mL buffer A supplemented with 10, 20, and 30 mM imidazole, and CrSBPase was eluted in three steps with 10 mL buffer A supplemented with 100, 200, and 300 mM imidazole. Pooled eluates containing CrSBPase were loaded on HiLoad Superdex 200 26/600 size-exclusion chromatography column and eluted in buffer A. Protein purity was assessed by electrophoresis on 12% acrylamide gel under denaturing and reducing conditions (supplementary figure 1). Peak fractions were assembled and concentrated by ultrafiltration on Amicon units of MWCO 30,000 Da cut-off. Final concentrations of purified proteins were in the 1-10 mg/mL range as measured by Nanodrop ND-2000 spectrophotometer using an ϵ_{280} of 24,722.5 M⁻¹.cm⁻¹.

Western blot

Polyclonal antibodies were generated by immunization of rabbits with pure recombinant CrSBPase (Genecust, Boynes France). Analyzed fractions were separated by denaturing polyacrylamide electrophoresis and transferred to 0.2 µm nitrocellulose membrane for detection with primary antibody subsequently detected by secondary anti-rabbit antibodies coupled to horseradish peroxidase. Detection was done with commercial ECL peroxidase assay (GE Healthcare, Chicago IL USA) with a Chemidoc (Bio-Rad, Hercules CA USA).

Crystallization and structure determination

Pure recombinant His6-CrSBPase-WT concentrated at 3-6 mg/mL was tested for crystallization on screens JCSG I-IV in 200 nL sitting drops. Single crystals were obtained and optimized with mother liquor 0.1 M sodium HEPES pH=7.5, 2% (v/v) polyethylene glycol 400, 2.0 M ammonium sulfate, flash-frozen in mother liquor supplemented with 25% glycerol and tested for X-ray diffraction. Complete X-ray diffraction datasets were collected at SOLEIL beamline Proxima-2A and allowed the determination of the crystal structure by model replacement with 5iz3, the structure of ortholog SBPase from the moss *Physcomitrium patens*. Model building in COOT and refinement in the PHENIX software suite yielded structure deposited under PDB ID 7b2o (Emsley, Lohkamp et al. 2010 [DOI](#)) (Adams, Afonine et al. 2011 [DOI](#)) (Liebschner, Afonine et al. 2019 [DOI](#)). After we obtained our experimental structure, the European Bioinformatics Institute (Hinxton, UK) communicated high-accuracy prediction models computed by Deepmind ALPHAFOLD2 (Jumper, Evans et al. 2021 [DOI](#)), including a prediction of CrSBPase structure (<https://alphafold.ebi.ac.uk/entry/P46284> [DOI](#)) that matches chain A in 7b2o crystal structure with a root mean square deviation (RMSD) of 0.453 Å. Hence, X-ray crystallography and AlphaFold independently contribute to the proposed structural analysis.

In order to obtain the structure of CrSBPase in a reduced state, crystals were grown in the presence of 10 mM of the reducing agent tris-(2-carboxyethyl)phosphine (TCEP) in condition JCSG IV E8/56 0.2 M lithium sulfate, 0.1 M Tris pH 7.0, and 1.0 M sodium/potassium tartrate. Crystals were cryo-protected in mother liquor supplemented with 25% ethylene glycol, flash-frozen and tested for X-ray diffraction at SOLEIL beamline Proxima-2A. A complete dataset was collected that allowed the determination of the structure by molecular replacement with 7b2o as a template,

resulting in structure 7zuv. Among the eight CrSBPase copies of the asymmetric unit, two had their disulfides absent as a consequence of the TCEP treatment while the others six subunits were essentially identical to untreated structures in entry 7b2o (*i.e.*, fully oxidized form).

Enzymatic assays

The catalytic activity of 10-200 nM pure recombinant CrSBPase was spectrophotometrically assayed in buffer A by coupling the hydrolysis of 1 mM FBP into F6P to the reduction of 0.5 mM NADP⁺ into NADPH through isomerization of F6P into glucose-6-phosphate (G6P) by 0.5 U.mL⁻¹ phosphoglucose isomerase (PGI) and the oxidation of G6P into 6-phosphogluconolactone (6PGL) by 0.1-0.5 U.mL⁻¹ glucose-6-phosphate dehydrogenase (G6PDH). Reporter assay is coupled in a molar ratio FBP:F6P:G6P:6PGL of 1:1:1:1 to the oxidation of one molar equivalent of NADP⁺ into NADPH that we recorded over 20 minutes by measuring the absorbance at 340 nm using a UVIKON spectrophotometer. Pre-treatments with 0-100 mM reduced dithiothreitol (DTTred), 0-30 mM MgSO₄, and 0-15 μM recombinant CrTRX-f2 supplemented with 1 mM DTTred (Lemaire, Tedesco *et al.* 2018 [\[4\]](#)) were carried out for 30 min at room temperature before measuring enzyme activity as described above.

Titration of accessible cysteine thiols

Quantification of free reactive cysteines on recombinant CrSBPase was performed with 5,5'-dithio-bis-(2-nitrobenzoic acid) (DTNB, Ellman's reagent) mixed with the purified recombinant protein in a 200:1 molar ratio in a 1 mL reaction volume containing 100 mM Tris-HCl, pH 7.9 as described previously (Meloni, Fanti *et al.* 2024 [\[4\]](#)). Reaction of DTNB with solvent-exposed cysteine thiols produces 2-nitro-5-thiobenzoate (TNB⁻) that we quantified by monitoring the absorbance at 412 nm and a molar extinction coefficient of TNB⁻ of 14.15 mM⁻¹.cm⁻¹. We calculated the number of free reactive cysteines per CrSBPase monomer by dividing the molar concentration of TNB⁻ by that of the protein.

Molecular dynamics

Molecular dynamics (MD) simulations were performed with the Gromacs package (v2021.4). A dimer of CrSBPase was solvated in a rhombic dodecahedron box with at least 8 Å between the protein atoms and the box edges, which corresponds to ~64000 atoms in total (~18000 water molecules) and a triclinic box of 97x97x68 Å³. The box was neutralized with 18 sodium atoms to make it neutral. The starting conformation was the crystallized structure of oxidized CrSBPase, and we protonated C115 and C120 to manually create the reduced state. The protein was described with the Amber14SB force field and water with the TIP3P model. Non-bonding interactions were described with standard parameters: van der Waals interactions had a cut-off of 8 Å, and electrostatic interactions were computed with PME with default parameters with a separation between spaces at 8 Å. Bonds containing a hydrogen atom were restrained with the LINCS algorithm with default parameters. After an energy minimization, the system was equilibrated under the NPT conditions with a δt=1 fs timestep, the velocity-rescale thermostat and the Berendsen barostat. Velocities were generated at 100 K and the system was heated up to 300 K in 400 ps before performing 100 ps at 300 K. We then ran simulations of 20 μs under the NPT conditions with a δt=2 fs timestep, the velocity-rescale thermostat and the Parrinello-Rahman barostat.

Cultivation of *Chlamydomonas* for native SBPase analysis

Chlamydomonas reinhardtii strain D66 was grown in 50 mL Tris-acetate-phosphate (TAP) medium under 40 μE.m⁻².s⁻¹ continuous light until density reached 5x10⁴ cell.mL⁻¹, harvested by 10 min centrifugation at 3,000 g. Cell pellet was resuspended in 100 mM Tris-HCl, pH 7.9, 100 mM NaCl, 5 mM L-cystéine, 500 μM NADH, and 1 mM EDTA (buffer L) and lysed by 3 freeze-thaw cycles in liquid nitrogen and water bath. Soluble fraction was separated by 10 min centrifugation at 30,000 g and loaded for size-exclusion chromatography on a calibrated Superose 6 16/600 column

equilibrated in buffer L. Aliquots of 500 μ L elution fractions were analyzed by western blot with primary antibodies raised in rabbits against pure recombinant CrSBPase (Covalab, Bron, France) and secondary anti-rabbit IgG coupled to peroxidase (Sigma-Aldrich reference A9169, Saint Louis, USA) revealed by ECL Prime (GE Healthcare).

Small-angle X-ray scattering

Pure recombinant CrSBPase was injected on Agilent HPLC BioSEC-3 300 column in buffer A, in line with the X-ray beam capillary at SOLEIL beamline SWING. CrSBPase was untreated or pre-treated with 10 mM of reducing agent dithiothreitol (DTTred). Diffusion curves were analyzed with ATSAS software to determine the molecular mass and oligomeric state of the protein.

Results and Discussion

Crystal structure of CrSBPase

Pure recombinant CrSBPase crystallized in space group $P2_12_12$ and provided a complete X-ray diffraction dataset at a resolution of 3.09 Å. The asymmetric unit is composed of six polypeptide chains packing as three dimers and four water molecules could be modeled. All chains align within a root mean square deviation (RMSD) <0.25 Å and can be considered identical. 309 out of the 331 residues were built in the electron density, the unmodelled sequences are in the disordered extensions at the amino- and carboxy-termini of the model. CrSBPase folds into two distinct domains (**figure 1A-B**). The amino-terminal domain (NTD, residues 73-247) is composed of helices 1, 2, and 3 and a mixed β -sheet formed by strands 3-9. The carboxy-terminal domain (CTD, residues 248-383) is composed of helices 4, 5, and 8 and helices 6 and 7 that sandwich a mixed β -sheet formed by strands 10-13. α -helices 1-3 pack on the amino-terminal end of the seven-strand sheet, while α -helices 4, 5, and 8 pack on the carboxy-terminal end of the four-strand sheet. α -helices 6 and 7 are sandwiched in between the two sheets. CrSBPase chain A aligns with previously reported crystal structure of PpSBPase with an RMSD of 0.478 Å (Güttele et al. 2018 PDB 5IZ3). This close structural similarity between orthologues indicates common structural properties supporting a conservation of the SBPase function in the evolutionary time separating the microalga *Chlamydomonas* from the land moss *P. patens*. A short antiparallel β -sheet is constituted of 4-residue strands 1 and 2 separated by a β -turn and preceded by loop A₁₁₃SCAGTAC₁₂₀. This motif projects to solvent from α -helices 2 and 3 (**figure 1C-D**). Residues 113-130 forming the loop and β -hairpin (LBH) motif are positioned away from the core of the protein, at 15.3 Å from water molecule 401 (W401), and will be further discussed in the next sections.

AF2 entry AF-P46284-F1-model_v4 from AlphaFold Protein Structure Database aligns with our crystal structure 7b2o chain E with RMSD = 0.434 Å, showing excellent agreement between experiment and prediction at the level of protein main chain. Exceptions are in local differences in several loops conformations and in the N-ter and C-ter borders of secondary structure elements. Numerous amino acid residues side chains adopt moderately distinct orientations between the computational model and the experimental structure.

AF3 was recently communicated (Abramson, Adler et al. 2024) along with its online prediction server hosted at <https://golgi.sandbox.google.com/>. CrSBPase model from AF3 aligns to our crystal structure 7B2O chain A with RMSD = 0.489 Å showing again their strong similarity and with a small discrepancy between AF2 and AF3 models with RMSD = 0.216 Å. The only significant deviations between 7B2O and AF3 are in the orientation of several side chains and notably on the conformation of region 114-131 that contain the LBH motif.

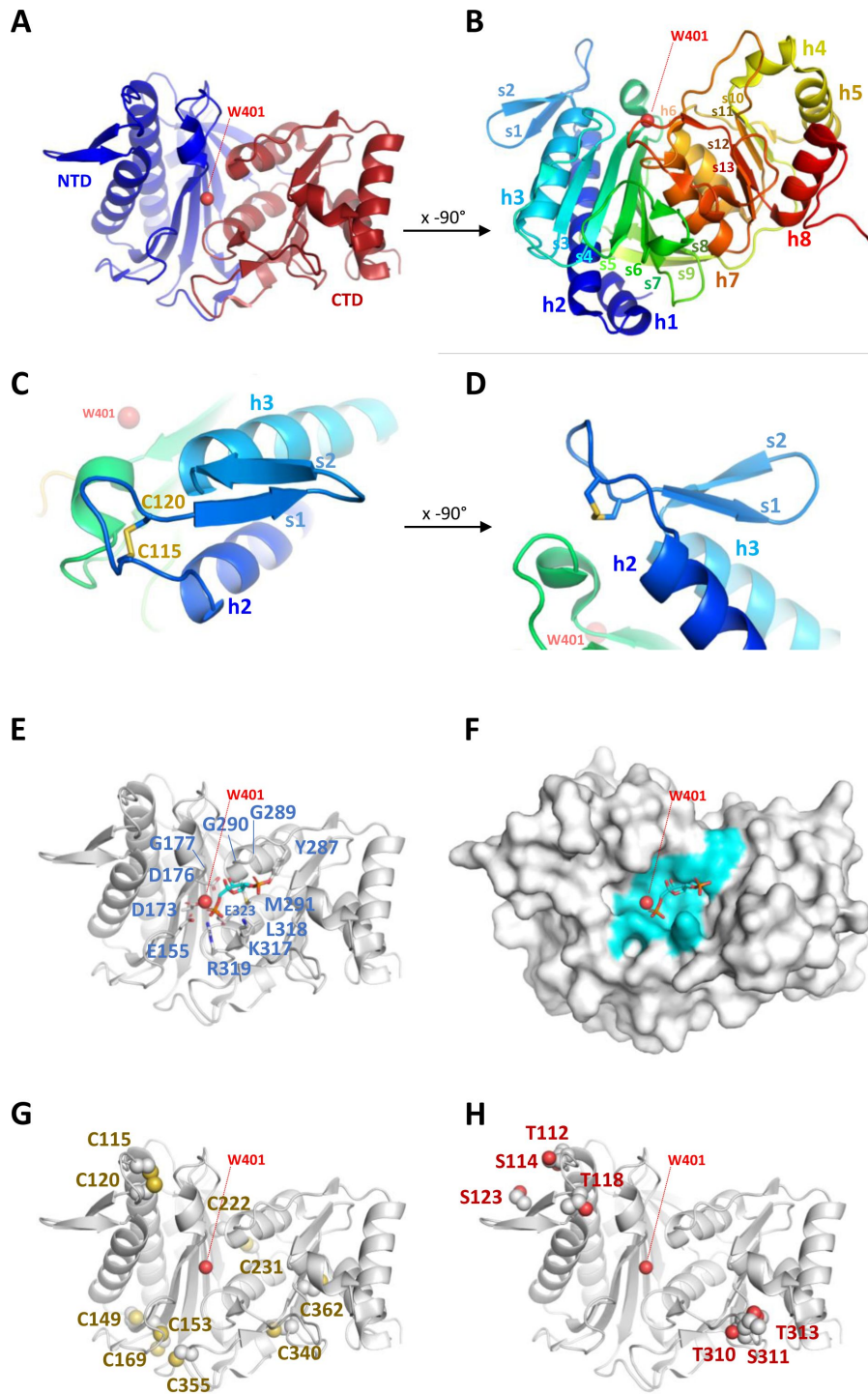


Figure 1.

Crystal structure of CrSBPase. (A) Topology is displayed as cartoon with main chain colored from blue (amino-terminus) to red (carboxy-terminus). (B) Rotated view of A. by x-axis over -90°. (C) Close-up view of region A₁₁₃SCAGTAC₁₂₀ with disulfide bond shown in sticks. (D) Rotated view of C. by x-axis over -90°. (E) Putative active site residues inferred from alignment with FBPase bound to FBP (5I0a, human muscle fructose-1,6-bisphosphatase E69Q mutant in active R-state in complex with fructose-1,6-bisphosphate) are represented in sticks. Residue numbering is according to Chlamydomonas SBPase Uniprot entry P46284. (F) Surface representation in the B. orientation with putative active site residues colored in cyan. Water molecule 401 (W401) oxygen is represented as a red sphere. (G) Cysteines side chains are represented in spheres. (H) Threonines and serines side chains reported to be the target of phosphorylations are represented in spheres.

CrSBPase catalytic pocket

CrSBPase is structurally similar to its ortholog from the moss *Physcomitrium patens* (RMSD=0.478 Å) (Gutle, Roret et al. 2016 [\[1\]](#)). SBPase is a homolog of FBPase, another enzyme involved in the CBBC in photosynthetic organisms and also ubiquitously involved in neoglucogenesis. Cytosolic isoforms of FBPase from non-plant sources have been extensively studied (Moorhead, Hodgson et al. 1994 [\[2\]](#), Lee and Hahn 2003 [\[3\]](#)). For instance, the crystal structure of human muscle FBPase complexed with its substrate (FBP) was solved (PDB entry: 5l0a, to be published). We aligned the structure CrSBPase onto this human FBPase:FBP complex from entry 5l0a, confirming their similarity (RMSD = 0.824 Å) with the notable exception of the LBH motif which is absent in FBPase. In a 4-Å distance to FBPase co-crystallized with FBP we aligned 12 residues that are likely available for FBP or SBP binding by CrSBPase (**figure 1E** [\[4\]](#)): E155, D173, D176, G177, Y287, G289, G290, M291, K317, L318, R319, and E323. These residues are exposed to the solvent in a continuous surface (**figure 1F** [\[4\]](#)). W401 could be modelled at the center of the putative SBPase catalytic pocket, in the vicinity of E155, D173, D176, R319, and E323, in the catalytic pocket predicted from alignment with human FBPase bound to FBP. W401 specifically is in bonding distance to catalytic residues D178, D181, and E328. W401 is likely to hydrolyze the substrate SBP.

Sites of redox post-translational modifications

Ten cysteines could be located in each monomer of CrSBPase (C115, C120, C149, C153, C169, C222, C231, C340, C355, C362). Previous redox-based proteomic studies exploiting *Chlamydomonas* cells or protein extract established that CrSBPase undergoes redox modifications by thioredoxin-dependent dithiol/disulfide exchange (Lemaire, Guillon et al. 2004 [\[5\]](#), Pérez-Pérez, Mauriès et al. 2017 [\[6\]](#)), S-glutathionylation (Zaffagnini, Bedhomme et al. 2012 [\[7\]](#)), or S-nitrosylation (Morisse, Zaffagnini et al. 2014 [\[8\]](#)). While the putative Cys sites for S-glutathionylation are not known, thioredoxin and S-nitrosylation Cys targets were identified as being C222/C231/C355/C362, and C355/C362, respectively. DTNB-based thiol titration on the native purified SBPase indicates that 4.0 ± 0.6 cysteines are likely reactive owing to their relative exposure to solvent in a free thiol form. Amino acid residue accessible surface area and accessibility (ASA) calculation is consistent with DTNB titration, with C149, C153, C231, and C355 exposed to the solvent (ASA respective areas: 0.154 Å², 0.113 Å², 0.308 Å², and 10.083 Å²), and the six others either shielded from solvent (C169, C222, C340, and C362) or engaged in a disulfide bridge (C115, C120) (Ahmad, Gromiha et al. 2004 [\[9\]](#)). In CrSBPase crystal structure, C222 and C231 are positioned on neighboring strands of the amino-terminal domain, exposing their thiol 3.5 Å away from each other (**figure 1G** [\[4\]](#), supplementary figures 2A-B). The formation of an intramolecular disulfide bridge at this cysteines pair is possible if local main chain rearrangements allow a closer contact. C362 thiol is positioned 10.5 Å away from the nearest thiol group of C340. Additionally, C340 and C362 are exposed on an opposite side of the carboxy-terminal β-sheet, disfavoring the formation of an internal disulfide bridge. C355 points its side chain towards C153 and C169 with a thiol-thiol distance of 4.6 Å and 10.1 Å, respectively. Although not observed in our model, a C355-C153/C169 dithiol-disulfide exchange is thus plausible, as described for plant FBPase (Jacquot, Lopez-Jaramillo et al. 1997 [\[10\]](#)). Local rearrangements that occur upon FBPase disulfide reduction causes a larger scale relaxation of the active site required to reach maximal catalytic potential (Chiadmi, Navaza et al. 1999 [\[11\]](#)) as kinetically characterized for enzymes from *Arabidopsis thaliana* (Yoshida and Hisabori 2018 [\[12\]](#)) and *Nicotiana benthamiana* (Zimmer, Swart et al. 2021 [\[13\]](#)). In *Chlamydomonas* and *Physcomitrium* SBPases, the FBPase-type regulatory disulfide is not observed but the SBPase-type C115-C120 bridge is conserved in a nearly identical position in the LBH motif where it may regulate the enzyme by alternative conformational rearrangements. Indeed, the disulfide-bonded cysteines constrain *Chlamydomonas* SBPase loop A₁₁₃SCAGTAC₁₂₀ into a compact lasso that could open into a relaxed conformation hypothetically activating catalysis.

Sites of post-translational phosphorylations

CrSBPase was reported to be phosphorylated from algae extracts at residues T112, S114, T118, S123, T310, S311, and T313 (Wagner, Gessner et al. 2006 [↗](#), Werth, McConnell et al. 2017 [↗](#), McConnell, Werth et al. 2018 [↗](#)). Consistently with the proximity of these residues in two sectors of the sequence (supplementary figure 3), the candidate modification sites are located in two contiguous regions in the three-dimensional structure (**figure 1H** [↗](#)). T112, S114, T118, and S123 are exposed to the solvent of the amino-terminal domain. The residues belong to or are located nearby the A₁₁₃SCAGTAC₁₂₀ motif where phosphorylations may cross-signal with redox modifications on C115 and C120. T310, S311, and T313 are located on the other edge of the enzyme, at the solvent exposed tip of the carboxy-terminal domain. Their co-localization at the T₃₁₀SPT₃₁₃ motif likely facilitates a coordinated phosphorylation of the three residues by the same kinase. Because both groups of potential phospho-sites are located at 17-18 Å from the catalytic pocket (W401 taken as a reference point), there is no straightforward mechanism by which such modifications would exert a control over *CrSBPase* activity. Analysis of phospho-mimicking mutants along with the identification of the actual kinases/phosphatases couples that control *CrSBPase* phosphorylation state would allow to comprehend the role of this putative regulatory mechanism.

CrSBPase enzymatic activity

Recombinant *CrSBPase* was assayed for its capacity to catalyze FBP hydrolysis into F6P (Gutle, Roret et al. 2016 [↗](#)), a proxy for the hydrolysis of SBP into S7P that we could not test by lack of available substrate and kinetic reporter method. Previous studies demonstrated that plant SBPase requires the cofactor Mg²⁺ and a chemical reductant (DTTred) to be fully activated *in vitro*, reaching a K_M value of 0.05 mM and a k_{cat} of 81 sec⁻¹ in optimal conditions (Anderson 1974 [↗](#), Woodrow and Walker 1980 [↗](#), Cadet and Meunier 1988 [↗](#)). In agreement with this, we found a total absence of activity in the untreated *CrSBPase* (data not shown), whereas pre-incubation of the enzyme with DTTred and Mg²⁺ resulted in a strong activation (**figure 2** [↗](#)). By testing different concentrations of MgSO₄ (3, 10, 15, and 20 mM) and DTTred (1, 10, 25, 50, and 100 mM), we established that 10 mM of DTT and 10 mM MgSO₄ are the optimal conditions to attain the maximum specific activity (**figure 2A** [↗](#) and 2B). Unexpectedly, DTTred concentrations above 10 mM had an inhibitory effect, causing a drastic drop in activity that probably resulted from structural effects leading to denaturation of the enzyme (**figure 2A** [↗](#)). In contrast, we observed no such inhibitory effect in the presence of magnesium at concentrations above 10 mM (**figure 2B** [↗](#)). Under physiological conditions, the light-dependent redox regulation of plant SBPase is specifically exerted by f-class thioredoxins (Gutle, Roret et al. 2016 [↗](#)), which are reduced by the ferredoxin-thioredoxin cascade (Schürmann and Buchanan 1975 [↗](#)). The chloroplast TRX-f2 from *Chlamydomonas reinhardtii* was thus tested and validated for its capacity to activate *CrSBPase* by reduction (**figure 2C** [↗](#)). The maximum activity, corresponding to 12.15 ± 2.15 μmol(NADPH) min⁻¹ mg(SBPase)⁻¹, was obtained by pre-incubating the enzyme with 1 μM *CrTRX-f2* supplemented with 1 mM DTT and 10 mM MgSO₄, and maintained even at higher *CrTRX-f2* concentrations (i.e., 5 and 10 μM) (**figure 2C** [↗](#)). *CrSBPase* specific activity that we report is comparable to that reported for spinach SBPase with FBP as substrate (3.5 μmol min⁻¹ mg⁻¹ (Seuter, Busch et al. 2002 [↗](#))) and to that reported for another recombinant *Chlamydomonas* SBPase with SBP as substrate (7.15 ± 0.61 μmol min⁻¹ mg⁻¹ (Tamoi, Nagaoka et al. 2005 [↗](#))).

Redox controlled dynamics of *CrSBPase*

CrSBPase crystallized under reducing treatment (10 mM TCEP). Among the eight subunits in the asymmetric unit, C115-C120 disulfide is absent in two chains (subunits C colored in magenta and subunits E colored in salmon, **figure 3D** [↗](#) and 3F), while the other six subunits still have the C115-C120 disulfide bridge (**figure 3B** [↗](#), C,E,G-I). All other cysteines are unaffected by the reducing treatment, consistently with their thiol form in crystal structure of oxidized state. These six

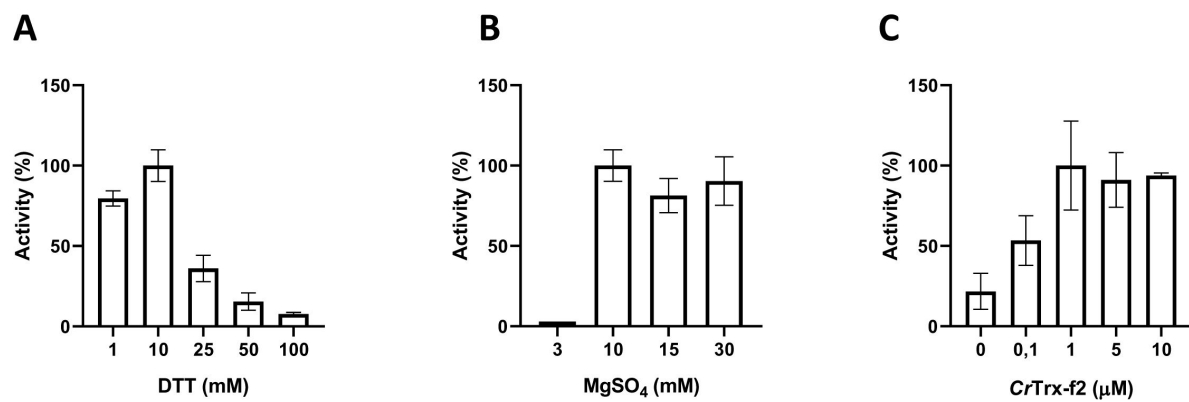


Figure 2.

Functional characterization of CrSBPase *in vitro*. Reported enzymatic activity of CrSBPase assayed for **(A)** reduction by reduced dithiothreitol (DTTred) and 10 mM of MgSO₄, **(B)** Magnesium sulfate (MgSO₄) and 10 mM DTTred, and **(C)** recombinant thioredoxin f2 from *Chlamydomonas reinhardtii* (CrTRX-f2) and 10 mM MgSO₄, 1 mM DTT.

subunits present mildly variable conformations of the A₁₁₃SCAGTAC₁₂₀ lasso conformation similar to the untreated oxidized structure of CrSBPase (**figure 3A**). Reduction of the C115-C120 disulfide increases local disorder, and the main chain electron density became uninterpretable over a few residues for subunits C (residues 115-118, 124-127), and subunits E (residues 119-127) (supplementary figure 4).

Molecular dynamics (MD) simulations of the reduced structure were performed by starting from a dimer of the crystallized oxidized form and forcing C115 and C120 residues to be reduced in the thiol state (-SH). We then ran two independent replicas of 20 μ s of MD. In addition, we also ran two independent replicas of 2 μ s of MD in the oxidized form. In one of the replicas of the reduced form (MD1), the protein structure barely changed as observed in the SG(C115)/SG(C120) distance that plateaued at ~ 7 Å, as well as in the RMSD with respect to the starting conformation that stayed around 2 Å (see supplementary figures 8-9 for plots and 12 for images of the structures). However, in the other replica of the reduced form (MD2), significant changes of conformation appeared: the SG(C115)/SG(C120) distance ended up at ~ 10 Å with peaks at 20 Å and the RMSD values ended up at 3.7 and 2.8 Å (supplementary figures 8-9). The difference between MD1 and MD2 is stronger when one focuses on the RMSD of the mobile motif (residues 112 to 131, supplementary figure 10). We observed that after 17 μ s of MD all chains adopted an equilibrated conformation and then fluctuated around that conformation (supplementary figure 10); representative structures extracted from MD were thus obtained from a clustering on the time frame (or block) 17-20 μ s.

The two chains A and B from MD2 adopt a different final conformation (see **figure 4A** for the overlap with the crystallized oxidized structure and **figure 4B** for the overlap with the crystallized reduced structure), and we cannot directly conclude on which one is the most stable. Moreover, the residues 115 to 128 of each chain from MD1/2 do not overlap well with the crystallized reduced structure which means that neither of the two chains conformations from MD simulations has converged towards the crystallized state. However, we can observe in the reduced crystallized structure that residues 134-148 form an α -helix which is also present in chain B from MD2 whereas this structure is not kept in chain A from MD2 (see green chain on the right of **figures 4A-B**). In addition, mobile motif from chain B seems to open towards the solvent in a similar way with what was observed for FBPase (Chiadmi, Navaza et al. 1999). Thus, we conclude that chain B from MD2 is more representative of the true reduced conformation than chain A from MD2. If we compare the mobility of chain B from MD2 in the reduced state with one chain in the oxidized state (also chain B from MD2), we can see that the mobile motif is overall more flexible in the reduced state, which is especially true for residues around C120 (residues 118 to 123, supplementary figure 11). We then analyzed the RMSD of residues 112 to 131 from chain B during MD2 (supplementary figure 12): residues 112, 130 and 131 barely move which is expected since they belong or are directly connected to the two α -helices flanking the mobile motif. At ~ 6 μ s of the trajectory almost all other residues (113-129 residues with the sole exception of C120) display an abrupt change of conformation, with a different magnitude though: starting from ~ 2 Å, the RMSD jumped to 3 Å for some residues (residue 129) and to 15 Å for other residues (residues 125, 126) (supplementary figures 9, 10, and 12). At ~ 13.5 μ s, another change occurs which involves C120 but not all the other residues. Thus, we conclude that the opening of the mobile motif happens in a concerted way and not sequentially residue per residue.

Oligomeric states of CrSBPase

Previous studies reported a homodimeric state for SBPase from the moss *Physcomitrella patens*, deduced from the quaternary structure of recombinant protein in the asymmetric unit of the crystal (Güttele, Roret et al. 2016). Size-exclusion chromatography (SEC) profile of CrSBPase is polydisperse with at least three distinct oligomeric states (**figure 6A**). Notably, we observed that the protein elutes as a main peak of 79 kDa, close to the mass of a homodimer (theoretical mass from monomer sequence: 36,176.7 Da) and consistently with the crystallized state of moss enzyme. Thorough inspection of the elution profile of CrSBPase revealed a peak broadened by a later-

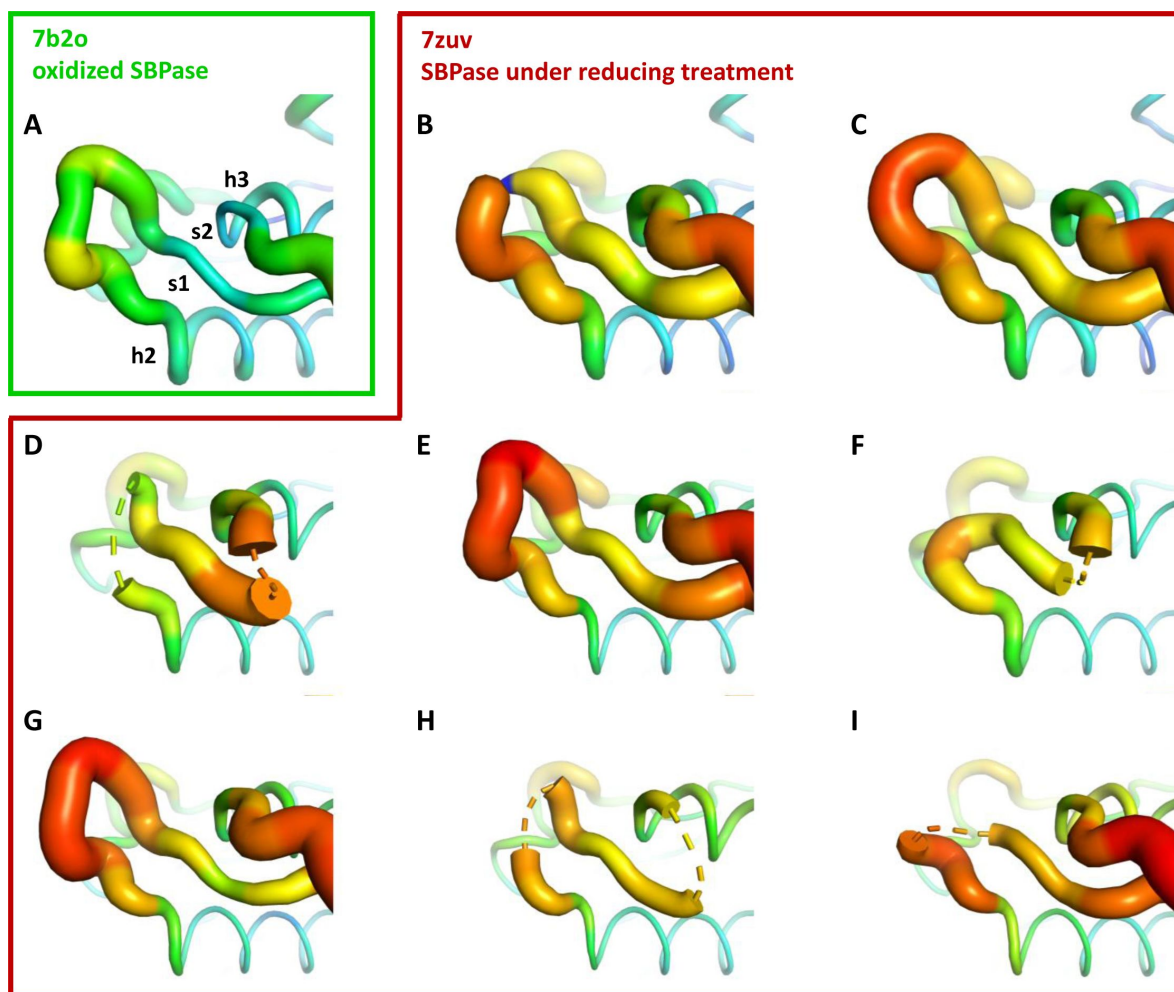


Figure 3.

Crystallographic structures of *CrSBPase* under reducing treatment: local disorder of the A₁₁₃SCAGTAC₁₂₀ loop. Main chain was traced according to crystallographic b-factor, with large orange sections representing high b-factor values and thin blue sections representing low b-factors. A-I. Aligned structures of *CrSBPase* protomers without redox treatment (A, 7b2o chain A) or in the presence of 10 mM TCEP reducing agent (B-I, 7zuv chains A-H).

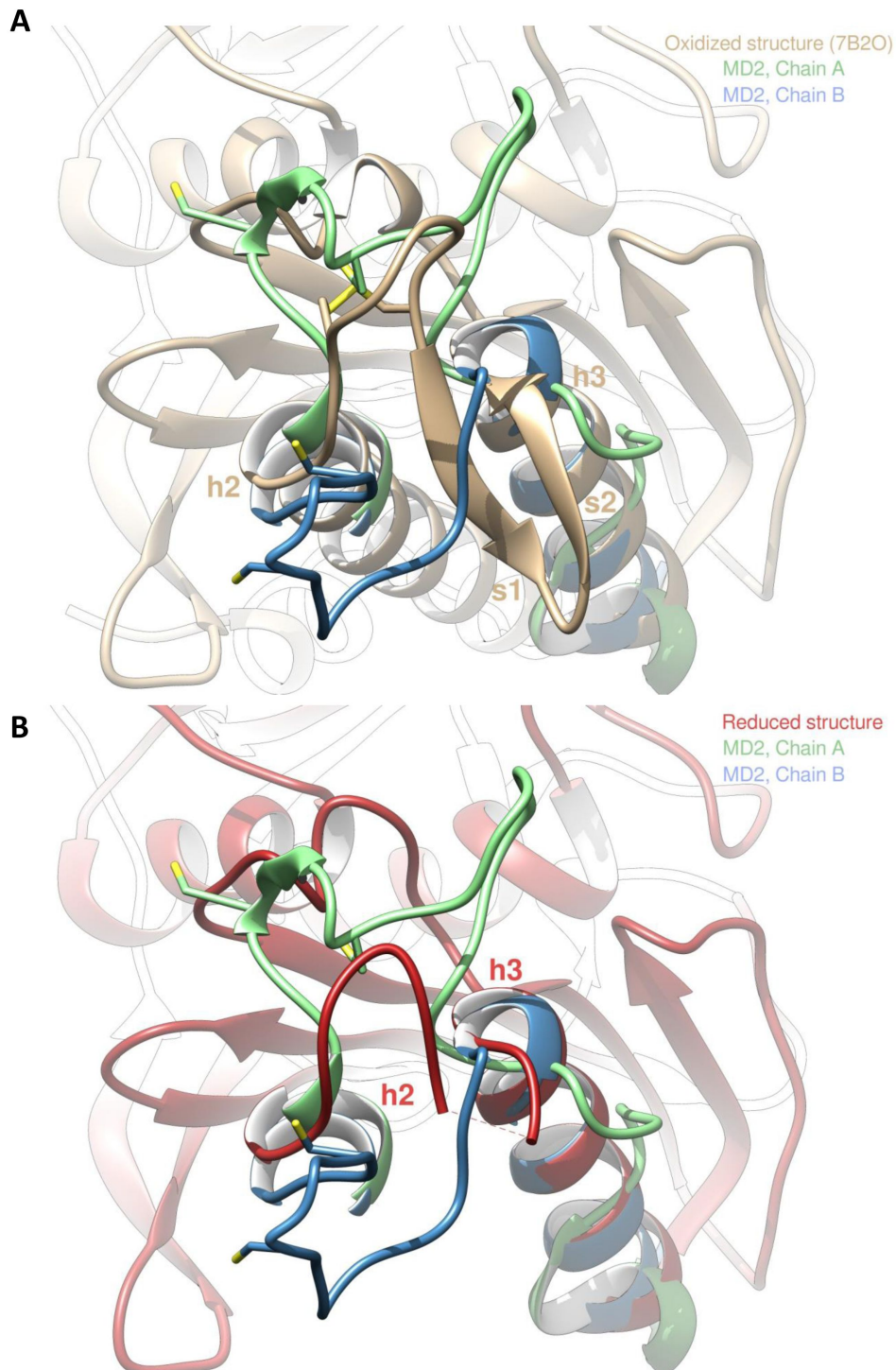


Figure 4.

Molecular dynamics simulation of *CrSBPase* after reduction. **(A)** Overlap of the crystallographic structure of oxidized *CrSBPase* and representative structures of equilibrated reduced *CrSBPase* during molecular dynamics simulation 2 (MD2). For structures extracted from MD, only residues 109 to 148 are displayed since most of the other residues are closely overlapping those of the crystallographic structure. **(B)** Overlap of the crystallographic structure of reduced *CrSBPase* and representative structures of equilibrated reduced *CrSBPase* during MD2. For structures extracted from MD, only residues 109 to 148 are displayed.

eluting shoulder, with the apparent molecular mass of the tailing species at 48 kDa suggesting the presence of a monomeric state of *CrSBPase* *in vitro*. A third *CrSBPase* species elutes at an apparent molecular mass of 145 kDa, that fits with a homotetramer. According to the absorbance of each peak and considering that they represent all states of purified and oxidized *CrSBPase* in solution, we estimate a repartition percentage of species of 10:53:37 for the tetramer:dimer:monomer mixture (total = 100%). Re-injection of the dimer peak fraction over size-exclusion chromatography yielded a similar mixed profile, supporting a dynamic equilibrium over the time of the experiment (>2 h) (data not shown).

CrSBPase was crystallized in space groups $P2_12_12$ or $P12_11$ with respectively 6 and 8 protomers in the asymmetric unit. Proteins Interfaces Structures and Assemblies (PISA) analysis reveals that protomers of both crystals pack as homodimers with extensive subunit interface of buried area 4,170 Å² in a similar manner (**figure 5A**) (Krissinel and Henrick 2007). In the asymmetric unit of *CrSBPase* crystallized in reducing conditions (7zuv) or within crystallographic neighboring units of *CrSBPase* in oxidized state (7b2o), *CrSBPase* further packs into a homotetramer, a dimer of dimers (**figure 5B**). The apparent molecular mass of SBPase extracted from *Chlamydomonas* cultures cultivated in TAP medium under 40 μE.m⁻².s⁻¹ illumination is comprised between 130 and 40 kDa with the homodimer representing the predominant state as supported by immunoblot analysis (**figure 6B**).

To further investigate the oligomeric dynamics of *CrSBPase*, we performed small-angle X-ray scattering (SAXS) analysis of purified (untreated) or reduced enzyme *in vitro* which reveals an increase in the protein radius of gyration from 39 to 62 Å upon reduction (supplementary figure 6). This observation implies a change in the oligomeric state of *CrSBPase* correlating with the redox state. As a matter of fact, purified *CrSBPase* tends to precipitate into amorphous aggregates that can be turned into droplets of liquid-liquid phase separation by treatment with 10 mM TCEP (supplementary figure 7). Since the LBH redox module is positioned at the homodimer interface (**figure 5C**), we postulated that the reduction of the C115-C120 disulfide bond and the subsequent local conformation changes depicted by MD affect the oligomer equilibrium of the enzyme. As a matter of fact, we observed that protein mutants C115S or C120S, which are unable to form the C115-C120 disulfide bond, elute in a different oligomer ratio than wild-type *CrSBPase* (**figure 6A**). Indeed, we observed a relative accumulation of monomer and tetramer species at the expense of the dimer state in both mutants. This suggests that C115-C120 is involved in the stabilization of the dimer. Meanwhile, the activity of the two Cys mutants still responded to DTT/Mg-dependent activation resembling the response of the wild-type protein. No activity was detected for native and untreated forms, while specific activities of C115S and C120S mutants upon treatment with DTT and MgSO₄ (10 mM for both compounds) were respectively 96% and 107% with respect to wild-type maximum activity (13.3 ± 2.5 and 14.9 ± 0.9 μmol(NADPH) min⁻¹ mg(SBPase)⁻¹ for C115S and C120S mutants, respectively, supplementary figure 14). Based on these data, we can thus conclude that C115-C120 disulfide bond is not the sole thiol-dependent regulatory mechanism involved in the modulation of *CrSBPase* activity and that the redox dependency of *CrSBPase* catalysis likely depends on other cysteine pairs. Future studies are required to shed light on the identity of cysteine residues that contribute to *CrSBPase* catalytic modulation other than C115 and C120.

Conclusions and perspectives

We report the first crystal structure of an SBPase from the microalga *Chlamydomonas reinhardtii*, a model for the molecular and cell biology of plants. SBPase is a photosynthetic enzyme involved in the regeneration phase of the CBBC cycle proved to be target of multiple regulations that we map onto our structural model. The native folding of *CrSBPase* is highly similar to that of SBPase from the moss *Physcomitrium patens* and to FBPase from chloroplast or cytoplasmic origins. Notwithstanding its propensity to form crystals, recombinant *CrSBPase* encompasses a range of

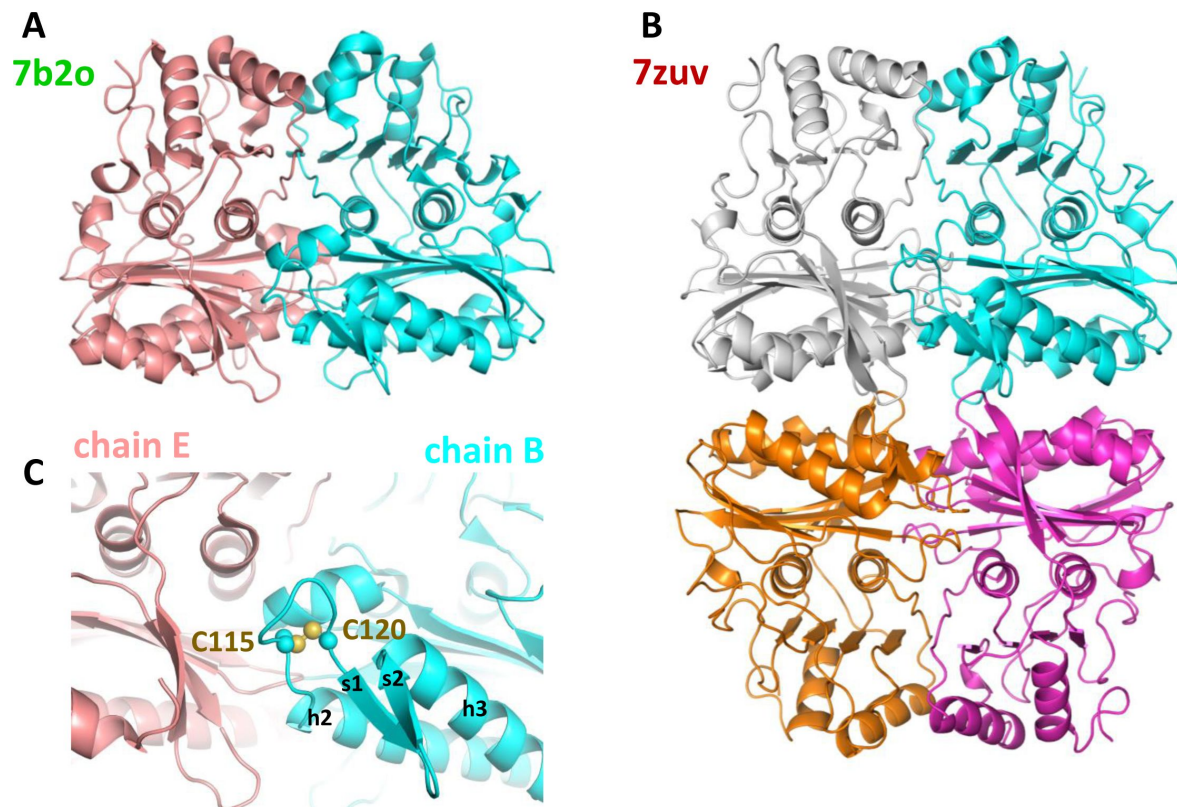


Figure 5.

Oligomeric state of *CrSBPase* in the crystal. **(A)** Asymmetric unit dimer of untreated *CrSBPase* (7b2o). Chains are represented in cartoon and colored cyan (chain B) and salmon (chain E). **(B)** Asymmetric unit homotetramer under reducing treatment (7zuv). Chains A (cyan), C (magenta), F (white), and H (orange) belong to the same asymmetric unit. **(C)** Close-up view on figure A homodimer interface. Loop 113-ASCAGTAC-120 from chain B (in cyan) is in 5-Å distance of neighboring chain E (in salmon). C115 and C120 are bonded by a disulfide bridge.

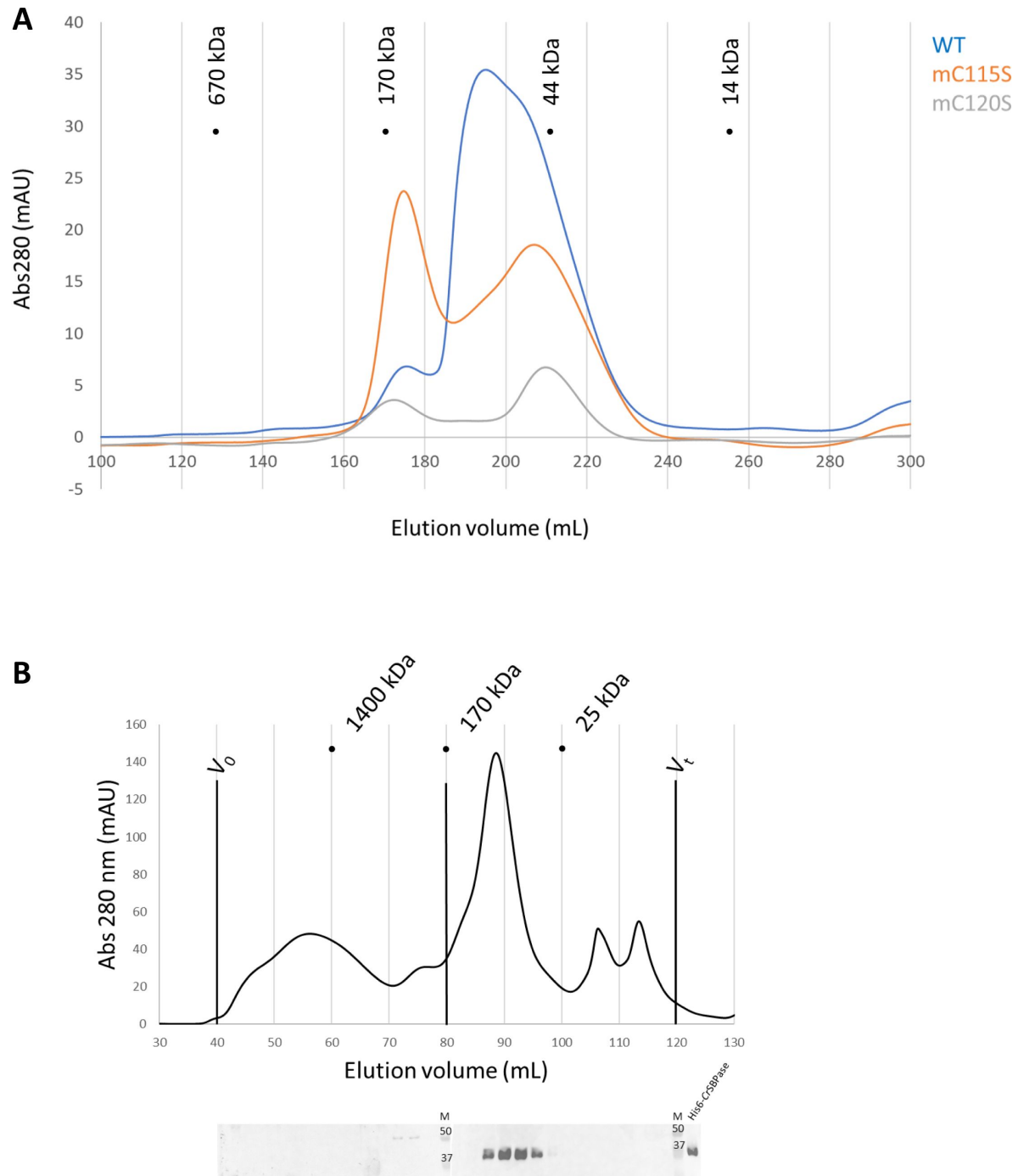


Figure 6.

Oligomeric state of *CrSBPase* *in vitro*. **(A)** Size-exclusion chromatograms of *CrSBPase* wild-type (blue), mutant C115S (orange), mutant C120S (gray) on Superdex 200 26/600 GL column. **(B)** Size-exclusion fractionation of *Chlamydomonas* cell extracts. *Chlamydomonas* cell culture was harvested, lysed and the soluble fraction of the lysate was loaded on Superose6 16/600 size-exclusion column. Chromatography fractions were analyzed by western blot with anti-*CrSBPase* primary antibodies. First membrane was loaded with fractions eluted from 40 to 80 mL. Second membrane was loaded with fractions eluted from 80 to 120 mL. M lane is loaded with molecular mass standards ladder. Recombinant *CrSBPase* was loaded on the last lane to the right.

oligomeric states, dominated by a homodimeric form susceptible to exchange for monomers, tetramers and higher-order assemblies. This oligomer equilibrium is dynamic and correlates with the presence of a disulfide bridge at the vicinity of the monomer-monomer interface, in the SBPase-specific LBH motif. We postulate that the reversible reduction/oxidation of the C115-C120 pair modulates the oligomerization of CrSBPase, through the induction of a local disorder that we observed in the crystal structure of the protein under reducing conditions and that correlates with computations of molecular dynamics.

How this oligomeric exchange impacts the actual activity of the enzyme is an open question but should be aligned with previous reports on the allosteric character of the structurally similar FBPase enzyme which follows dimer/tetramer exchanges (Barciszewski, Wisniewski et al. 2016 [↗](#)) with a Hill cooperativity coefficient close to 2 representing positive cooperativity in the context of a homodimer (Giudici-Orticoni, Buc et al. 1990 [↗](#)). Recent *in vivo* proteome mapping in *Chlamydomonas reinhardtii* localized SBPase in a punctate region surrounding the Rubisco pyrenoid nearby five other CBBC enzymes: phosphoglycerate kinase 1 (PGK), glyceraldehyde-3-phosphate dehydrogenases (GAP1, GAP3), fructose-bisphosphate aldolase (FBA3), ribulose-5-phosphate 3-epimerase (RPE1), and phosphoribulokinase (PRK) (Wang, Patena et al. 2022 [↗](#)). Based on our structure-function study on recombinant SBPase, we can speculate that the altered redox state of SBPase may influence the movement to or from the liquid partition of the stroma, where SBPase is metabolically required to provide its contribution to CBBC reactions near other enzymes of the pathway.

Authors contributions

JH, NC: Conceptualization; JH, NC, MS, TLM: Data curation; JH, NC, MZ: Formal analysis; JH: Funding acquisition; JH, NC, MZ, SDL, MS, TLM: Investigation; JH, NC, MZ, MS, TLM: Methodology; JH, SDL: Project administration; JH, NC, SDL: Resources; NC: Software; JH, MZ: Supervision; JH, NC, MZ, SDL, MS, TLM: Writing - original draft; JH, NC, MZ, SDL, MS, TLM: Writing - review & editing.

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Conflicts of interest statement

The authors report no conflict of interest relevant to the conduct or report of this research.

Table 1.

Crystallographic data collections and models building statistics.

	7b2o (untreated)	7zuv (partially reduced)
Wavelength (Å)	0.9801	0.9801
Resolution range (Å)	46.62 - 3.095 (3.206 - 3.095)	48.54 - 3.11 (3.221 - 3.11)
Space group	P 2 ₁ 2 ₁ 2	P 1 2 ₁ 1
Unit cell (Å, °)	178.224 183.652 75.196 90 90 90	53.774 163.462 172.765 90 91.939 90
Total reflections	615155 (55989)	376091 (32376)
Unique reflections	45740 (4245)	53456 (5213)
Multiplicity	13.4 (12.9)	7.0 (6.2)
Completeness (%)	99.34 (94.20)	99.65 (97.17)
Mean I/sigma(I)	12.02 (1.72)	8.54 (1.30)
Wilson B-factor (Å ²)	64.82	68.45
R-merge	0.495 (1.877)	0.2492 (1.502)
R-meas	0.5144 (1.954)	0.2691 (1.64)
R-pim	0.1386 (0.5343)	0.1009 (0.6477)
CC1/2	0.983 (0.623)	0.988 (0.479)
CC*	0.996 (0.876)	0.997 (0.805)
Reflections used in refinement	45607 (4242)	53417 (5193)
Reflections used for R-free	1986 (183)	1983 (191)
R-work	0.1942 (0.3071)	0.1963 (0.3239)
R-free	0.2390 (0.3482)	0.2328 (0.3740)
CC(work)	0.944 (0.788)	0.956 (0.726)
CC(free)	0.912 (0.649)	0.949 (0.596)
Number of non-hydrogen atoms	14194	18714
Macromolecules	14190	18704
Solvent	4	10
Protein residues	1859	2451
RMS(bonds) (Å)	0.005	0.004
RMS(angles) (°)	0.75	0.73
Ramachandran favored (%)	94.26	94.80
Ramachandran allowed (%)	5.52	4.70
Ramachandran outliers (%)	0.22	0.50
Rotamer outliers (%)	0.13	0.55
Clashscore	4.88	6.60
Average B-factor (Å ²)	67.00	68.41
macromolecules (Å ²)	67.01	68.40
solvent (Å ²)	49.59	94.37

Statistics for the highest-resolution shell are shown in parentheses.

Supplementary table 1.

Primers used for point mutagenesis.

5'-forward C115S	CCGCACCGCCTCGAGCGCCGGTACCGCCTGCGTG
3'-reverse C115S	CACGCAGGCGGTACCGGCGCTCGAGGCGGTGCGG
5'-forward C120S	GCGCCGGTACCGCCAGCGTGAACAGCTTCGGCG
3'-reverse C120S	CGCCGAAGCTGTTCACGCTGGCGGTACCGGCGC

References

1. Abramson J. *et al.* (2024) **Accurate structure prediction of biomolecular interactions with AlphaFold 3** *Nature*
2. Adams P. D. *et al.* (2011) **The Phenix software for automated determination of macromolecular structures** *Methods* **55**:94–106
3. Ahmad S., Gromiha M., Fawareh H., Sarai A. (2004) **ASAView: database and tool for solvent accessibility representation in proteins** *BMC Bioinformatics* **5**
4. Almagro Armenteros J. J., Salvatore M., Emanuelsson O., Winther O., von Heijne G., Elofsson A., Nielsen H. (2019) **Detecting sequence signals in targeting peptides using deep learning** *Life Sci Alliance* **2**
5. Anderson L. E (1974) **Activation of pea leaf chloroplast sedoheptulose 1,7-diphosphate phosphatase by light and dithiothreitol** *Biochem Biophys Res Commun* **59**:907–913
6. Barciszewski J., Wisniewski J., Kolodziejczyk R., Jaskolski M., Rakus D., Dzugaj A. (2016) **T-to-R switch of muscle fructose-1,6-bisphosphatase involves fundamental changes of secondary and quaternary structure** *Acta Crystallogr D Struct Biol* **72**:536–550
7. Cadet F., Meunier J. C. (1988) **pH and kinetic studies of chloroplast sedoheptulose-1,7-bisphosphatase from spinach (*Spinacia oleracea*)** *Biochem J* **253**:249–254
8. Calvin M (1962) **The path of carbon in photosynthesis** *Science* **135**:879–889
9. Chiadmi M., Navaza A., Miginiac-Maslow M., Jacquot J. P., Cherfils J. (1999) **Redox signalling in the chloroplast: structure of oxidized pea fructose-1,6-bisphosphate phosphatase** *EMBO J* **18**:6809–6815
10. Cotton C. A., Kabasakal B. V., Miah N. A., Murray J. W. (2015) **Structure of the dual-function fructose-1,6/sedoheptulose-1,7-bisphosphatase from *Thermosynechococcus elongatus* bound with sedoheptulose-7-phosphate** *Acta Crystallogr F Struct Biol Commun* **71**:1341–1345
11. Emanuelsson O., Brunak S., von Heijne G., Nielsen H. (2007) **Locating proteins in the cell using TargetP, SignalP and related tools** *Nat Protoc* **2**:953–971
12. Emsley P., Lohkamp B., Scott W. G., Cowtan K. (2010) **Features and development of Coot** *Acta Crystallogr D Biol Crystallogr* **66**:486–501
13. Feng L., Han Y., Liu G., An B., Yang J., Yang G., Li Y., Zhu Y. (2007) **Overexpression of sedoheptulose-1,7-bisphosphatase enhances photosynthesis and growth under salt stress in transgenic rice plants** *Funct Plant Biol* **34**:822–834
14. Feng L., Wang K., Li Y., Tan Y., Kong J., Li H., Li Y., Zhu Y. (2007) **Overexpression of SBPase enhances photosynthesis against high temperature stress in transgenic rice plants** *Plant Cell Rep* **26**:1635–1646

15. Franke D. *et al.* (2017) **ATSAS 2.8: a comprehensive data analysis suite for small-angle scattering from macromolecular solutions** *J Appl Crystallogr* **50**:1212–1225
16. Giudici-Orticoni M. T., Buc J., Bidaud M., Ricard J. (1990) **Thermodynamics of information transfer between subunits in oligomeric enzymes and kinetic cooperativity. 3. Information transfer between the subunits of chloroplast fructose biphosphatase** *Eur J Biochem* **194**:483–490
17. Glaser F., Pupko T., Paz I., Bell R. E., Bechor-Shental D., Martz E., Ben-Tal N. (2003) **ConSurf: identification of functional regions in proteins by surface-mapping of phylogenetic information** *Bioinformatics* **19**:163–164
18. Gutle D. D. *et al.* (2016) **Chloroplast FB Pase and SB Pase are thioredoxin-linked enzymes with similar architecture but different evolutionary histories** *Proc Natl Acad Sci U S A* **113**:6779–6784
19. Gütle D. D. *et al.* (2016) **Chloroplast FB Pase and SB Pase are thioredoxin-linked enzymes with similar architecture but different evolutionary histories** *Proc Natl Acad Sci U S A* **113**:6779–6784
20. Hammel A., Sommer F., Zimmer D., Stitt M., Mühlhaus T., Schroda M. (2020) **Overexpression of Sedoheptulose-1,7-Bisphosphatase Enhances Photosynthesis in Chlamydomonas reinhardtii and Has No Effect on the Abundance of Other Calvin-Benson Cycle Enzymes** *Front Plant Sci* **11**
21. Jacquot J. P., Lopez-Jaramillo J., Miginiac-Maslow M., Lemaire S., Cherfils J., Chueca A., Lopez-Gorge J. (1997) **Cysteine-153 is required for redox regulation of pea chloroplast fructose-1,6-bisphosphatase** *FEBS Lett* **401**:143–147
22. Johnson M. P (2016) **Photosynthesis** *Essays Biochem* **60**:255–273
23. Johnson X., Alric J. (2013) **Central carbon metabolism and electron transport in Chlamydomonas reinhardtii: metabolic constraints for carbon partitioning between oil and starch** *Eukaryot Cell* **12**:776–793
24. Jumper J. *et al.* (2021) **Highly accurate protein structure prediction with AlphaFold** *Nature* **596**:583–589
25. Krissinel E., Henrick K. (2007) **Inference of macromolecular assemblies from crystalline state** *J Mol Biol* **372**:774–797
26. Kubis A., Bar-Even A. (2019) **Synthetic biology approaches for improving photosynthesis** *J Exp Bot* **70**:1425–1433
27. Lee S. W., Hahn T. R. (2003) **Light-regulated differential expression of pea chloroplast and cytosolic fructose-1,6-bisphosphatases** *Plant Cell Rep* **21**:611–618
28. Lefebvre S., Lawson T., Zakhleniuk O. V., Lloyd J. C., Raines C. A., Fryer M. (2005) **Increased sedoheptulose-1,7-bisphosphatase activity in transgenic tobacco plants stimulates photosynthesis and growth from an early stage in development** *Plant Physiol* **138**:451–460
29. Lemaire S. D., Guillon B., Le Maréchal P., Keryer E., Miginiac-Maslow M., Decottignies P. (2004) **New thioredoxin targets in the unicellular photosynthetic eukaryote Chlamydomonas reinhardtii** *Proc Natl Acad Sci U S A* **101**:7475–7480

30. Lemaire S. D., Tedesco D., Crozet P., Michelet L., Fermani S., Zaffagnini M., Henri J. (2018) **Crystal Structure of Chloroplastic Thioredoxin f2 from Chlamydomonas reinhardtii Reveals Distinct Surface Properties** *Antioxidants* **7**
31. Liebschner D. *et al.* (2019) **Macromolecular structure determination using X-rays, neutrons and electrons: recent developments in Phenix** *Acta Crystallogr D Struct Biol* **75**:861–877
32. McConnell E. W., Werth E. G., Hicks L. M. (2018) **The phosphorylated redox proteome of Chlamydomonas reinhardtii: Revealing novel means for regulation of protein structure and function** *Redox Biol* **17**:35–46
33. Meloni M., Fanti S., Tedesco D., Gurrieri L., Trost P., Fermani S., Lemaire S. D., Zaffagnini M., Henri J. (2024) **Characterization of chloroplast ribulose-5-phosphate-3-epimerase from the microalga Chlamydomonas reinhardtii** *Plant Physiol* **194**:2263–2277
34. Mettler T. *et al.* (2014) **Systems Analysis of the Response of Photosynthesis, Metabolism, and Growth to an Increase in Irradiance in the Photosynthetic Model Organism Chlamydomonas reinhardtii** *Plant Cell* **26**:2310–2350
35. Moorhead G. B., Hodgson R. J., Plaxton W. C. (1994) **Copurification of cytosolic fructose-1,6-bisphosphatase and cytosolic aldolase from endosperm of germinating castor oil seeds** *Arch Biochem Biophys* **312**:326–335
36. Morisse S., Zaffagnini M., Gao X. H., Lemaire S. D., Marchand C. H. (2014) **Insight into protein S-nitrosylation in Chlamydomonas reinhardtii** *Antioxid Redox Signal* **21**:1271–1284
37. Nishizawa A. N., Buchanan B. B. (1981) **Enzyme regulation in C4 photosynthesis. Purification and properties of thioredoxin-linked fructose bisphosphatase and sedoheptulose bisphosphatase from corn leaves** *J Biol Chem* **256**:6119–6126
38. Ohta J (2022) **A novel variant of the Calvin-Benson cycle bypassing fructose bisphosphate** *Sci Rep* **12**
39. Pérez-Pérez M. E., Mauriès A., Maes A., Tourasse N. J., Hamon M., Lemaire S. D., Marchand C. H. (2017) **The Deep Thioredoxome in Chlamydomonas reinhardtii: New Insights into Redox Regulation** *Mol Plant* **10**:1107–1125
40. Raines C. A (2022) **Improving plant productivity by re-tuning the regeneration of RuBP in the Calvin-Benson-Bassham cycle** *New Phytol* **236**:350–356
41. Robert X., Gouet P. (2014) **Deciphering key features in protein structures with the new ENDscript server** *Nucleic Acids Research* **42**:W320–W324
42. Rosenthal D. M., Locke A. M., Khozaei M., Raines C. A., Long S. P., Ort D. R. (2011) **Over-expressing the C(3) photosynthesis cycle enzyme Sedoheptulose-1-7 Bisphosphatase improves photosynthetic carbon gain and yield under fully open air CO(2) fumigation (FACE)** *BMC Plant Biol* **11**
43. Schmollinger S. *et al.* (2014) **Nitrogen-Sparing Mechanisms in Chlamydomonas Affect the Transcriptome, the Proteome, and Photosynthetic Metabolism** *Plant Cell* **26**:1410–1435

44. Schürmann P., Buchanan B. B. (1975) **Role of ferredoxin in the activation of sedoheptulose biphosphatase in isolated chloroplasts** *Biochim Biophys Acta* **376**:189–192
45. Seuter A., Busch M., Hain R. (2002) **Overexpression of the potential herbicide target sedoheptulose-1,7-bisphosphatase from *Spinacia oleracea* in transgenic tobacco** *Molecular Breeding* **9**:53–61
46. Simkin A. J., Lopez-Calcano P. E., Davey P. A., Headland L. R., Lawson T., Timm S., Bauwe H., Raines C. A. (2017) **Simultaneous stimulation of sedoheptulose 1,7-bisphosphatase, fructose 1,6-bisphosphate aldolase and the photorespiratory glycine decarboxylase-H protein increases CO₂ assimilation, vegetative biomass and seed yield in *Arabidopsis*** *Plant Biotechnol J* **15**:805–816
47. Tamoi M., Nagaoka M., Shigeoka S. (2005) **Immunological properties of sedoheptulose-1,7-bisphosphatase from *Chlamydomonas* sp. W80** *Biosci Biotechnol Biochem* **69**:848–851
48. Wagner V., Gessner G., Heiland I., Kaminski M., Hawat S., Scheffler K., Mittag M. (2006) **Analysis of the phosphoproteome of *Chlamydomonas reinhardtii* provides new insights into various cellular pathways** *Eukaryot Cell* **5**:457–468
49. Wang L. *et al.* (2022) **A Chloroplast Protein Atlas Reveals Novel Structures and Spatial Organization of Biosynthetic Pathways** *bioRxiv*
50. Werth E. G., McConnell E. W., Gilbert T. S., Couso Lianez I., Perez C. A., Manley C. K., Graves L. M., Umen J. G., Hicks L. M. (2017) **Probing the global kinome and phosphoproteome in *Chlamydomonas reinhardtii* via sequential enrichment and quantitative proteomics** *Plant J* **89**:416–426
51. Woodrow I. E (1982) **Sedoheptulose-1,7-bisphosphatase from wheat chloroplasts** *Methods Enzymol* **90**:392–396
52. Woodrow I. E., Walker D. A. (1980) **Light-mediated activation of stromal sedoheptulose bisphosphatase** *Biochem J* **191**:845–849
53. Yoshida K., Hisabori T. (2018) **Determining the Rate-Limiting Step for Light-Responsive Redox Regulation in Chloroplasts** *Antioxidants* **7**
54. Zaffagnini M., Bedhomme M., Marchand C. H., Morisse S., Trost P., Lemaire S. D. (2012) **Redox regulation in photosynthetic organisms: focus on glutathionylation** *Antioxid Redox Signal* **16**:567–586
55. Zimmer D. *et al.* (2021) **Topology of the redox network during induction of photosynthesis as revealed by time-resolved proteomics in tobacco** *Sci Adv* **7**

Editors

Reviewing Editor

Shozeb Haider

University College London, London, United Kingdom

Senior Editor

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University of Freiburg, Freiburg, Germany

Reviewer #1 (Public Review):

In this study, Le Moigne and coworkers shed light on the structural details of the Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae *Chlamydomonas reinhardtii*. The SBPase is part of the Calvin cycle and catalyzes the dephosphorylation of sedoheptulose-1,7-bisphosphate (SBP), which is a crucial step in the regeneration of ribulose-1,5-bisphosphate (RuBP), the substrate for Rubisco. The authors determine the crystal structure of the CrSBPase in an untreated, oxidized state. Based on this structure, potential active site residues and sites of post-translational modifications are identified. Furthermore, the authors determine the CrSBPase structure in a reduced state revealing the disruption of a disulfide bond in close proximity to the dimer interface. The authors then use molecular dynamics (MD) to gain insights into the redox-controlled dynamics of the CrSBPase and investigate the oligomerization of the protein using small-angle X-ray scattering (SAXS) and size-exclusion chromatography. Despite the difference in oligomerization, disruption of this disulfide bond did not impact the activity of CrSBPase, suggesting additional thiol-dependent regulatory mechanisms modulating the activity of the CrSBPase.

The authors provide interesting new findings on a redox-mechanism that modulates the oligomeric behavior of the SBPase. Comparisons of the *Chlamydomonas* structure to the previously determined SBPase structure from the moss *Physcomitrium patens* confirm a high structural similarity between the two proteins suggesting that this mechanism might be evolutionary conserved. Future research will have to address this question experimentally, also considering potential cooperativity between the subunits to confirm the link between oligomerization and SBPase activity.

<https://doi.org/10.7554/eLife.87196.2.sa2>

Reviewer #2 (Public Review):

The central theme of the manuscript is to report on the structure of SBPase - an enzyme central to the photosynthetic Calvin-Benson-Bassham cycle. The authors claim that the structure is the first of its kind from a chlorophyte *Chlamydomonas reinhardtii*, a model unicellular green microalga. The authors use a number of methods like protein expression, purification, enzymatic assays, SAXS, molecular dynamics simulations and x-ray crystallography to resolve a 3.09 Å crystal structure of the oxidized and partially reduced state. The results are supported by the claims made in the manuscript. While the structure is the first from a chlorophyte, it is not unique. Several structures of SBPase are available and a comparison has been made between the structure reported here and others that have been previously published.

<https://doi.org/10.7554/eLife.87196.2.sa1>

Author response:

The following is the authors' response to the original reviews.

eLife assessment

The manuscript reports useful findings by resolving the crystal structure of Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae Chlamydomonas

reinhardtii, which is involved in the Calvin cycle. The data presented are solid based on validated methodologies, which help in understanding the structure and function of this enzyme.

We thank the editors for this positive assessment.

Public Reviews:

Reviewer #1 (Public Review):

In this study, Le Moigne and coworkers shed light on the structural details of the Sedoheptulose-1,7-Bisphosphatase (SBPase) from the green algae Chlamydomonas reinhardtii. The SBPase is part of the Calvin cycle and catalyzes the dephosphorylation of sedoheptulose-1,7-bisphosphate (SBP), which is a crucial step in the regeneration of ribulose-1,5-bisphosphate (RuBP), the substrate for Rubisco. The authors determine the crystal structure of the CrSBPase in an oxidized state. Based on this structure, potential active site residues and sites of post-translational modifications are identified. Furthermore, the authors determine the CrSBPase structure in a reduced state revealing the disruption of a disulfide bond in close proximity to the dimer interface. The authors then use molecular dynamics (MD) to gain insights into the redox-controlled dynamics of the CrSBPase and investigate the oligomerization of the protein using small-angle X-ray scattering (SAXS) and size-exclusion chromatography. Despite the difference in oligomerization, disruption of this disulfide bond did not impact the activity of CrSBPase, suggesting additional thiol-dependent regulatory mechanisms modulating the activity of the CrSBPase.

We thank reviewer 1 for his/her careful reading of our manuscript.

The authors provide interesting new findings on a redox-mechanism that modulates the oligomeric behavior of the SBPase, however without investigating this potential mechanism in more detail. The conclusions of this manuscript are mostly supported by the data, but they should be more carefully evaluated in respect to what is known from other systems as e.g. the moss Physcomitrella patens. This is especially of interest, as SBPase was previously reported to be dimeric, whereas for FBPase a dimer/tetramer equilibrium has been observed.

We thank reviewer 1 for his/her comments on the novel or confirmatory character of our structure-function analysis on CrSBPase. We address the questions of oligomeric states later in this response.

(1) Given that PpSBPase has been already characterized in detail, the authors should provide a more rigorous comparison to the existing data on SBPases. This includes a more conclusive structural comparison but also the enzymatic assays should be compared to the findings from P. patens. Do the authors observe differences between the moss and the chlorophyte systems, maybe even in regard to the oligomerization of the SBPase?

Indeed, a previous study conducted by one of the authors of the current manuscript (Stéphane D. Lemaire) and collaborators determined the structure and regulatory properties of SBPase from the moss *Physcomitrella patens* (Gütle *et al.* 2018 <https://doi.org/10.1073/pnas.1606241113>). We added a clearer reference to this earlier work. The differences that we observed regarding the oligomeric states of SBPase from *Chlamydomonas reinhardtii* principally stem from our analytical method *in vitro* through size-exclusion chromatography, in comparison with crystal packing analysis in the reference study. We detailed PpSBPase/CrSBPase oligomeric state comparison in the paragraph 'Oligomeric states of

CrSBPase'. Besides, the asymmetric unit of our CrSBPase crystal structure is also a homodimer, similarly to PpSBPase, and we suggest that PpSBPase is also likely to adopt several oligomeric states *in vitro*. If this were confirmed by experiments, SBPase in several organisms would behave analogously to FBPase regarding the dimer/tetramer equilibrium.

In paragraph 'Crystal structure of CrSBPase' we added a comparison by alignment of our CrSBPase crystal structure to the previously reported _Pp_SBPase crystal structure, stating that with RMSD=0.478 Å the proteins are essentially identical.

In paragraph 'CrSBPase enzymatic activity' we compared the value we obtained for enzyme specific activity to those previously published on other SBPase from Chlamydomonas or the land plant *Spinacia oleracea*, highlighting the similarity of results in three different systems and teams (Seuter *et al.* 2002 <https://doi.org/10.1023/A:1019297521424> and Tamoi *et al.* 2005 DOI: 10.1271/bbb.69.848).

(2) The authors should include the control experiments (untreated SBPase) and the assays performed with mutant versions of the SBPase, which are currently only mentioned in the text or not shown at all.

We add supplementary figure 14 in order to illustrate that since SBPase C115S or C120S mutants are still activated by reducing agent, the disulfide bridge between cysteines 115 and 120 is not the single control over SBPase activity but rather a control over the oligomeric exchange of the enzyme indirectly contributing to redox activation of the active site.

(3) The representation of the structure in figures (especially Figures 1 and 3) should be adjusted to match the author's statements. In Figure 1, the angle from which the structure is displayed changes over the entire figure making it difficult to follow especially as a non-structural biologist. Furthermore, important aspects of the structure mentioned in the text are not labeled and should be highlighted, by e.g. a close-up. Same holds true for Figure 3 that currently mostly shows redundant information.

We thank reviewer 1 for his/her advise on how to improve Figure 1. We drew new images for the complete figure, hopefully providing more consistent and clearer visual support to our text. For simplicity, protein is now always represented centered around its active site in the same orientation. We represent co-crystallized water in all projections as a guide to the eye.

Figure 3 and supplementary figure 3 were switched in order to better represent the experimental evidence provided by the resolution of SBPase structure under reducing conditions, i.e., the increase in local disorder around C115-C120 pair of cysteines in the 113-130 stretch forming a redox-conditionally dynamic loop and β -hairpin motif.

(4) The authors state that mutation of C115 and C120 to serine destabilize the dimer formation, while more tetramer and monomer is formed. As the tetramer is essentially a dimer of dimers, the authors should elaborate how this might work mechanistically. In my opinion, dimer formation is a prerequisite for tetramer formation and the two mutations rather stabilize the tetramer instead of destabilizing the dimer.

Time-dependent dynamic character of SBPase oligomer exchange is not resolved by the current study because we essentially combined size-exclusion chromatography (SEC) and X-ray crystallography to define quaternary structures at equilibrium. Overall, homodimer is the dominant state of wild-type SBPase by abundance in the purified recombinant form and by forming the constitutive asymmetric unit in all crystal packings. Dimer is indeed present in the tetramer state, a dimer of dimers, as pertinently stated by reviewer 1.

This being recognized, we tried to explain the systematic co-elution of the principal dimeric form with an additional species of smaller size on SEC (supplementary figure 1, right-side shoulder of the peak), at the apparent mass of a monomer. When solving the crystal structures of SBPase we realized that the dimer interface is contributed by residues 113-130 forming a loop and β -hairpin motif. Notably, in this loop cysteine 115 (C115) maps at bonding distance of 3.9 Å of side chain of arginine 220 (R220) from dimer partner subunit. In loop 113-120, cysteine pair C115 and C120 are subject to redox switching between disulfide (closed) and dithiol (open) conformations, as shown in our structures 7B2O and 7ZUV, respectively. Given that the reduction of C115-C120 disulfide bridge correlates with a higher flexibility of this motif that contributes to dimer interface (figure S3), we hypothesized that reduction of SBPase would destabilize dimer state to the benefit of transitory monomer state, and indeed point mutagenesis of C115S or C120S caused a large modification of oligomer equilibrium in favour of the monomer (figure S1C).

Mechanistically, we suggest two scenarios for the tetramer formation: either monomers first interact as in the crystallographic dimer before pairing such dimers into tetramers (as proposed by reviewer 1), or monomers start tetramerization by favoring the alternative subunit interface (figure 5B, between cyan and magenta chains) before stabilizing the crystallographic homodimer interface. In this latter case, monomerization would be necessary to efficiently re-arrange SBPase dimers into tetramers.

In physiological conditions the re-arrangement switch would be controlled by C115-C120 reduction through ferredoxin-thioredoxin redox cascade. Structural studies in dynamic conditions like native mass spectroscopy/photometry would be necessary to solve this speculation unambiguously although at this stage of our investigation there seem little doubt to us that C115-C120 disulfide-dithiol exchange is essential to control a dimer/monomer balance in first instance.

Reviewer #2 (Public Review):

*The central theme of the manuscript is to report on the structure of SBPase - an enzyme central to the photosynthetic Calvin-Benson-Bassham cycle. The authors claim that the structure is first of its kind from a chlorophyte *Chlamydomonas reinhardtii*, a model unicellular green microalga. The authors use a number of methods like protein expression, purification, enzymatic assays, SAXS, molecular dynamics simulations and xray crystallography to resolve a 3.09 Å crystal structure of the oxidized and partially reduced state. The results are supported by the claims made in the manuscript. One of the main weakness of the work is the lack of wider discussion presented in the manuscript. While the structure is the first from a chlorophyte, it is not unique. Several structures of SBPase are available. As the manuscript currently reads, the wider context of SBPase structures available and comparisons between them is missing from the manuscript. Another important point is that the reported structure of crSBPase is 0.453Å away from the alphafold model. Though fleetingly mentioned in the methods section, it should be discussed to place it in the wider context.*

We thank reviewer 2 for his/her assessment of our manuscript. In response to his/her suggestion to better compare our SBPase structure from the model microalga *Chlamydomonas reinhardtii* to that of the ortholog from *Physcomitrium patens* previously reported by an author of this manuscript (Stéphane D. Lemaire) and collaborators (Güttele et al. 2018), we wish to point out that paragraph 3 of the introduction was dedicated to this reference along with a mention to related *Thermosynechococcus elongatus* dual function fructose-1,6-bisphosphatase sedoheptulose-1,7-bisphosphatase (F/SBPase). We nevertheless follow his/her suggestion to better detail comparison between chloroplastic SBPase structures

in the first result section 'Crystal structure of CrSBPase', consistently with response 1 to reviewer 1 (see above).

Regarding the integration of AlphaFold (AF) computational models in a general discussion about SBPase molecular structure, we wish to point out that our initial 7B2O crystallographic model of CrSBPase was deposited in PDB on 2020-11-27 before AlphaFold2 was available for the scientific community (Jumper *et al.* publication date is 15 July 2021).

AF2 entry AF-P46284-F1-model_v4 from AlphaFold Protein Structure Database aligns with our crystal structure 7B2O chain E with RMSD = 0.434 Å, showing excellent agreement between experiment and prediction at the level of protein main chain. It must still be pointed out that it is the AF2 model which is at 0.434 Å away from the experiment, and not the opposite. Exceptions of alignments are in local differences in several loops conformations and in the length of secondary structure elements. Many amino acid residues side chains adopt distinct orientations between the computational model and the experimental structure.

AF3 was recently communicated (Abramson *et al.* 2024) along with its online prediction server hosted at <https://golgi.sandbox.google.com>. CrSBPase model from AF3 align to our crystal structure 7B2O chain A with RMSD = 0.489 Å showing again their strong similarity and with a smaller discrepancy between AF2 and AF3 of RMSD = 0.216 Å. The only significant deviations between 7B2O and AF3 are in the orientation of several side chains and notably on the conformation of region 114-131 that contain the redox sensor motif.

We added the last two paragraphs to the revised version of the manuscript, after the results section presenting our crystallographic work.

Recommendations for the authors:

We made all recommended modifications as detail below.

Reviewer #1 (Recommendations For The Authors):

I have outlined a number of minor points below.

We addressed all minor points listed.

Line 220: The asymmetric unit only contains three dimers. The dimer of dimer or tetramer can only be reconstituted by displaying the symmetry mates.

We corrected our sentence for 'The asymmetric unit is composed of six polypeptide chains packing as three dimers'.

I also suggest that the authors separate the description of the asymmetric unit content from the modeled water molecules and rephrase e.g. „...and four water molecules could be modeled.”

We rephrased as suggested.

I appreciate that the authors uploaded the structure in advance of this article, which allowed to evaluate the quality of the structure. Although this does not add valuable information, I have identified several unmodeled blobs, which possibly also account for waters.

Unmodeled blobs were tentatively assigned to water but had to be removed during later refinements. We used Coot Validate tools 'Unmodelled blobs' and 'Check/Delete water' to progress towards the current optimal refinement statistics. We admit that the resolution of

the crystallographic dataset (3.09 Å) is limiting to reliably model mobile or less resolved elements like water molecules. Overall, we estimate that the functional elements of the structure are modeled to the best of our knowledge and with minimal subjectivity.

Line 222: Please write 309 instead of spelling the number.

We corrected for 309 instead of spelling the number.

Line 223: The structure representation in Figure 1A/B has to be improved. The authors might consider labeling the two domains & color them in two colors instead of the rainbow color coding. Furthermore, the 90{degree sign} rotation does not add much information. Here, turning the model in a different direction that allows to see the central b-sheet of domain 2 might be better suited. Furthermore, instead of describing b-strands first, followed by a-helices, I suggest describing which secondary structure elements form the two domains.

We improved Figure 1A as suggested while keeping Figure 2B with 90° rotation as rainbow color gradient in order to display with clarity the secondary structure content and connectivity. The orientation was tilted to better display the central β-sheet. This new version of Figure 1A/B should facilitate the text description of SBPase architecture that we amended as suggested.

Line 229: The information on A113-120 should be depicted in a closeup in Figure 1A.

We made a close-up view of sequence 113-120 as added figures 1C-D and modified the rest of the figure and legend accordingly.

Line 234: Please provide an r.m.s.d here.

We now provide r.m.s.d. for all structural alignments.

Line 242: Please introduce the domain labeling in Fig 1C to make it easier to track the exact region within SBP here. Is the residue numbering according to SBP or the human FBP?

Modified version of figure 1 now shows SBPase in the same orientation for panels A, E, F, G, H for simplicity. Domains labeling is indicated in panel A with NTD/CTD distinct colors as suggested. We explicated the position of W401 on all panels as a guide to the eye. We indicated in figure legend that residue numbering is according to Chlamydomonas SBPase Uniprot entry P46284.

Line 244: Is Figure 1D in the same orientation as C? I suggest making the surface transparent and showing the cartoon below, which will allow to easier see the solvent accessibility of the residues. Also, clearly label W401 (although it's the only water shown/modeled in this region).

We modified figure 1 to show all equivalent panels (ie. A-E-F-G-H) with the same orientation. In this new form we think that solvent accessibility and the relative position of significant residues is easier to interpret for the reader. W401 is consistently labeled throughout figure 1 panels.

Line 263: Please provide a close-up of the C222 and C231 including measured distance. It's clearly not visible from this view. It might even be helpful to provide close-ups of all cysteine residues that are mentioned in the text.

In the modified version of figure 1 we estimate that C222 and C231 are more easily visible. We added a close-up view of C22-C231 environment in a new supplementary figure 2. Since we do not explore further the functional relevance of this redox pair we chose not include C222-C231 close-up view in main figure 1. We added legends and modified supplementary figures numbering accordingly.

Line 276: As already mentioned earlier, none of the panels in Figure 1 provide a close-up of this loop. This should be added.

This loop is now displayed as a close-up view in panels C and D of main figure 1.

Line 284: It is difficult to follow the relative positions of the potential modification sites if the model is always depicted from a different angle in Figure 1. The authors might want to change this across Figure 1 or show the rotation angle.

This problem was addressed in the revised figure 1, panels A-E-F-G-H are in the same orientation now. Panel B was kept at a rotation of 90° with corresponding annotation.

Line 290: Please label W401. Also stick to one nomenclature (W or H2O).

We labeled W401 and kept nomenclature consistent throughout the manuscript.

For comparative reasons, a full kinetic measurement (determination of K_m and k_{cat}) of the SBPase would also be helpful here.

We resolved to avoid a full kinetic measurement of CrSBPase because we could neither identify a reliable chemical provider nor synthesize ourselves the physiological substrate sedoheptulose-1,7-bisphosphate (SBP) and only characterized the reaction with fructose-1,6-bisphosphate. However, in the revised form of the manuscript we added in main text paragraph 'CrSBPase enzymatic activity' the kinetic constants from the previous reference study conducted on spinach SBPase (Cadet and Meunier, Biochem. J. 1988) with $K_{MSBP}=0.05$ mM and $k_{catSBP}=81$ sec⁻¹ of fully active enzyme with SBP as a substrate. For comparison, the authors of this study report that activity of SBPase on FBP is in the same range but lower, with $K_{MFBP}=0.38$ mM and $k_{catFBP}=21$ sec⁻¹. We also added a comparison of specific activities of our CrSBPase and spinach SBPase in the main text, showing that our enzyme behaves as previously reported ortholog from land plant.

Line 303: How much MgSO4 was used for the experiment shown in Figure 2A?

10 mM of MgSO4 was used for experiment shown in Figure 2A. We added this information in the figure legend. We also added in the legend that 10 mM DTT is present in the experiment of Figure 2B and that 10 mM of MgSO4 and 1 mM of DTT are present in the experiment of Figure 2C.

Line 321: In my opinion it is not necessary to show the regions of all molecules here. I was rather expecting a superposition of the two structures (oxidized and reduced) with a close-up of the respective disulfide in the two states.

We agree that the initial version of Figure 3 panels showing side-by-side all conformational variants of the redox motif appear redundant. We switched initial Figure 3 to supplementary data and replaced it with the crystallographic b-factor mapping of the redox motif, in the variable conditions resolved by the crystals. We would like to stress that all these conformations were experimentally determined through X-ray crystallography, whether of

the crystal of pure inactive enzyme that proved to be oxidized on the redox motif, or of the equivalent crystals submitted to activating treatment by the chemical reductant TCEP. As an attempt to clarification we added visual boxes to better appreciate this reduction-induced conformational plasticity that we interpreted as a local *conditional* disorder.

Line 331: Could the authors provide movies of the MD simulation? Otherwise, interpretation of the MD simulation results might be difficult for non-experts.

We added two movies of 20- μ sec MD simulations as supplementary data to help non-expert readers.

Line 343: It might be helpful to label the structure elements in Figure 4 accordingly (e.g. residues, etc.)

We added secondary structure labeling in Figure 4.

Line 381: Should be changed to Figure 5A.

We changed reference to figure 6 that is a renumbering of figure 5 with changes included from suggestions below. Figure 6 now includes chromatograms of recombinant SBPase in panel A and chromatogram and western blot analysis of Chlamydomonas extracts in panel B.

Line 383: See above, figure 5B. Which structure is shown in the figure? 7zuv or 7b2o? Maybe include both structures in the figure in a side-by-side view. The authors might also want to include the SEC chromatograms in the main figure. Especially the purification from Chlamydomonas is helpful to estimate whether post-translational modifications have an impact on the oligomerization. This should also be mentioned in the text.

7b2o and 7zuv are illustrated side-by-side in panels A and B of figure 5. This was indicated in the figure legend, we now added the information on the figure. As suggested above we included chromatograms initially presented as supplementary material in a new main figure 6, panel A for recombinant proteins and panel B for proteins extracted from Chlamydomonas. Initial figures 5D-E, showing surface conservation of the dimeric SBPase, is moved to supplementary figure 5.

Line 385: I don't find the cultivation of Chlamydomonas in the method section. It should be added.

We added a methods paragraph dedicated to « Cultivation of Chlamydomonas for native SBPase analysis ».

Line 390-392: This information is not really helpful. Concentrated purified proteins might precipitate after a week storage without physiologically relevant effects being the reason.

We agree that the observation of a precipitate building up *in vitro* after a week of storage bears no particular physiological implications. We rather intended to report that an aggregated form of purified protein can be turned to droplets under the redox conditions that activate the enzyme. We reformulated these lines for clarification.

Line 397: I would appreciate having the SEC-chromatograms of the mutants also in the main figure.

Size-exclusion chromatograms that were initially in supplementary figures are now shown in main text figure 6 panel A, with the profiles WT and mutants aligned.

Line 402: Where are these data shown? They should be included in Figure 5.

We added a figure to present these data, not shown in the initial version of the manuscript. We preferred to place it as supplementary material because C115S and C120S mutant catalytic activity is essentially the same as WT and do not reveal a direct mechanistic effect of C115-C120 reduction over the catalytic pocket.

Line 427: Did the authors look into a possible cooperativity of their SBPase?

We did not observe direct positive cooperativity that could be ascribed to allostery in our enzymatic assays. It was previously reported for spinach SBPase that SBP saturation functions were hyperbolic with no evidence of homotropic interactions in the enzyme oligomer (Cadet and Meunier *Biochem J.* 1988 253, 249-254). The authors of this kinetic study however present a clear sigmoid response of SBPase to Mg²⁺ concentration, suggestive of an activating cross-talk between active sites in the oligomer. We consider this hypothesis of interest and wish we could further investigate allosteric conformational changes when SBP physiological substrate would be available.

Line 428-434: I don't really understand how the proteome mapping fits in here. Do the authors speculate that SBPase is recruited by some of the identified enzymes or directly interacts with them or that rather the spatial distribution optimizes the reaction kinetics?

We indeed want to correlate our *in vitro* observations of CrSBPase conditions of activity to those recently published by the group of Dr. Martin Jonikas in a physiological, *in vivo* setup of *Chlamydomonas reinhardtii* (Wang, Patena *et al.* *Cell* 2023 186, 3499–3518). We have no experimental evidence demonstrating the first suggestion that SBPase is recruited or directly interacts with partner enzymes but we privilege the second suggestion that local spatial distribution in the chloroplast stroma optimizes enzyme reaction kinetic thanks to Calvin-Benson-Bassham enzymes proximity. We rephrased these lines to clarify our hypothesis and express its speculative character.

Reviewer #2 (Recommendations For The Authors):

To make the manuscript stronger, the authors are recommended to do the following:

We followed given recommendations.

(1) include a wider discussion on the other SBPase structures that are available. A detailed comparison should be made between the oxidized and reduced structures present in the PDB with the structures that are being reported in the manuscript.

Consistently with reviewer #1 suggestion, and as detailed in response to public review above, we followed the recommendation to better report previous structural studies of SBPase in the results section. We also added comparisons with computational models from AlphaFold2 and AlphaFold3.

(2) The authors mention co-operativity between the subunits. With excellent sampling from molecular dynamics simulations, the authors should demonstrate co-operativity between the subunits.

Our molecular dynamic (MD) simulations span 20 μ sec of SBPase in the dimeric state, starting from the experimental structures determined by XRC. In the considered time window, the only significant events that we observed are the local reorganization of the LBH motif that is

a prerequisite for dimer rearrangement. We infer that local disorder contributes a separation of the pair of subunits in order to later allow for the building of the active homotetramer, at longer time scales that are outside the capacities used in this work. Moreover, demonstrating cooperativity with MD simulations would require more than a single event to ensure that results are significant, and performing series of 20 μ s-MD of SBPase is also outside the available capacities.

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